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Housing Price Prediction Using Machine Learning

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

I have read the guidance to students on academic integrity, misconduct, and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project **module** or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6). I have not used ChatGPT, or any other generative AI tool, to write the report **or code (other than where I have** declared **or referenced it).**

I did not use human participants or undertake a survey in my MSc Project.

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UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

ACKNOWLEDGMENT

I extend my heartfelt gratitude to all those who have contributed to the successful completion of this research endeavor. First and foremost, I want to express my sincere appreciation to my Supervisor Klaas Wiersema whose guidance, support, and invaluable insights have been instrumental throughout this project.

Their expertise and encouragement have been indispensable in shaping the direction of my research and refining my methodologies. My gratitude also extends to the faculty and staff of the University of Hertfordshire, whose resources and facilities have facilitated the smooth progress of my research activities.

Lastly, I would like to express my heartfelt thanks to my family and friends for their unwavering support, understanding, and patience throughout this journey. This research would not have been possible without the collective efforts and support of all those mentioned above. Thank you for your invaluable contributions.

ABSTRACT

This report presents a comprehensive analysis of the "House Prices — Advanced Regression Techniques" dataset, aimed at predicting residential property sale prices. The dataset, sourced from Kaggle, consists of 1460 observations in the training set and 1459 in the test set, with 80 features covering various aspects of properties.

The analysis begins with an exploration of the dataset's characteristics through descriptive statistics and graphical representations, revealing insights into house prices, construction years, sales trends, and house sizes. Exploratory Data Analysis (EDA) uncovers patterns, trends, and relationships between features and sale prices, facilitating a deeper understanding of the dataset.

Following EDA, data preprocessing techniques are applied to clean, transform, and prepare the data for modeling. This includes handling missing values, encoding categorical variables, and feature engineering. Dimensionality reduction and data normalization techniques are also employed to improve model performance.

The report then proceeds to model building and evaluation, where various regression models are trained and evaluated using metrics such as Root Mean Squared Error (RMSE) and R-squared. Models including Ridge regression, Lasso regression, Support Vector Regression (SVR), and Random Forest are evaluated for their predictive accuracy. Visualization techniques such as scatter plots, and model performance comparisons are utilized to assess and interpret the models' behavior.

Overall, the analysis provides valuable insights into the factors influencing house prices and the performance of different regression models in predicting sale prices. The findings can inform stakeholders such as buyers, sellers, and real estate agents in making informed decisions related to residential property transactions.

ETHICAL CONSIDERATIONS

During this project, I have adhered to ethical principles and standards to ensure the responsible use of data and the ethical conduct of my analysis. I have respected the privacy and confidentiality of the individuals whose data is included in the "House Prices — Advanced Regression Techniques" dataset. This dataset doesn't include any personal information like names, email addresses, or phone numbers, and all the data has been anonymized and aggregated to prevent the identification of individuals. As the dataset used in this project is publicly available on Kaggle, it is assumed that the data has been shared with appropriate consent from the original contributors. I have not collected any new data or used any proprietary information without proper authorization.

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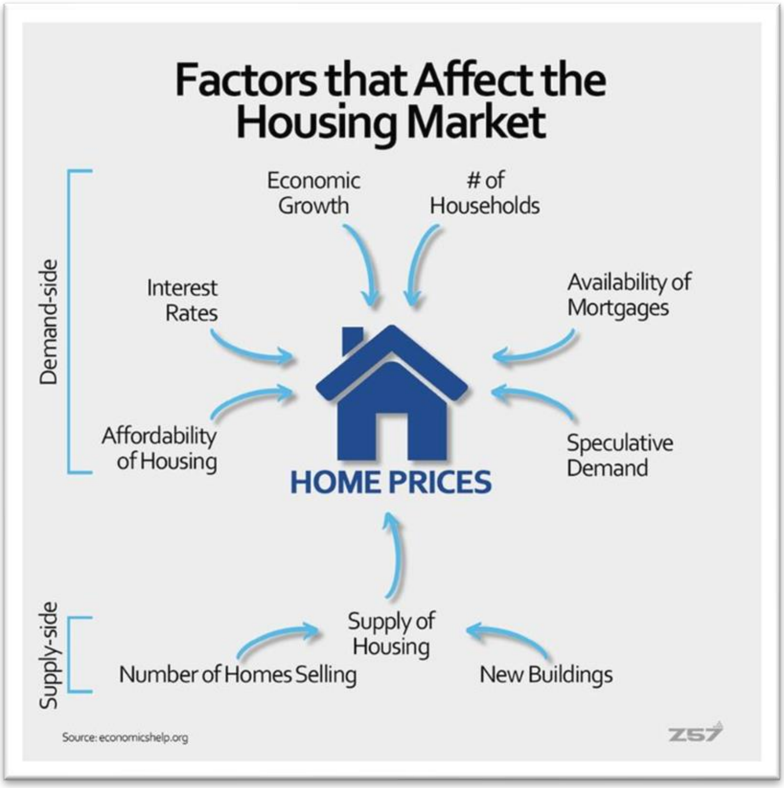
Chapter 1 – Introduction

1.1 Introduction:

The housing market plays a vital role in our economy, reflecting economic conditions and societal trends. Housing prices are affected by various factors like supply and demand, interest rates, demographics, and economic conditions. It's important for homebuyers, sellers, investors, and policymakers to understand these factors to predict housing prices accurately.[1]

Fig. 1 – Factors that affect the housing market



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In recent years, the housing market has gone through significant changes due to events like the 2008 financial crisis and the COVID-19 pandemic. These events have highlighted the need for better risk management tools in real estate. Advanced regression techniques provide a powerful way to analyze and predict housing prices, helping stakeholders make informed decisions about buying, selling, or investing in real estate. [2]

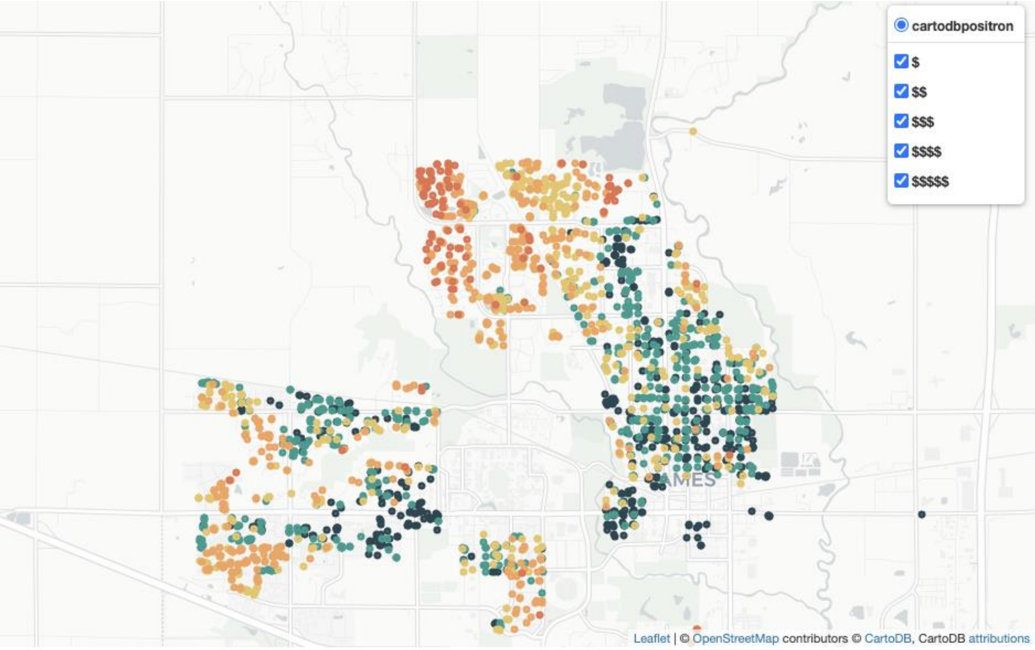
Advanced regression techniques provide a robust method for analyzing and predicting housing prices. By utilizing these techniques, researchers and analysts can identify key drivers of housing prices and create models that can forecast future price movements with precision. These predictive models are invaluable for stakeholders making decisions regarding real estate transactions.

1.2 Background:

Before diving into the details of the project, it is important to provide a brief overview of Ames, Iowa to gain further insight into the real estate market there. According to a 2010 USCB report, Ames had a population of about 59,000. The city's economy and demographics are based largely on Iowa State University, a public college in the heart of the city. With over 75% of the population being either college or Iowa State University students, Ames has the layout of a huge, expansive campus.

Interestingly, Iowa State University is the largest employer in Ames, accounting for 20% of all jobs in the city. As a result, Ames and other university towns have something in common: a sizable portion of their residences are rented, which contributes to the stability of the real estate market.

Fig. 2 - Ames Iowa Housing Price Distribution Map



https://imgur.com/a/cMi4m1b

The housing market in Ames reflects a common urban pattern, more affordable homes tend to be clustered around the city center and the Iowa State University campus, which is typical for areas with a high concentration of students and rental properties. On the other hand, the northern sections of Ames are characterized by higher-priced homes, indicating a preference for more exclusive residential areas away from the central hustle and bustle. This distribution creates distinct zones of economic housing diversity within the city.

1.3 Project Goals:

The objective of the project is to examine the patterns and determinants that influence in properties values in Ames, Iowa, utilizing the "House Price Advanced Regression Techniques" available on Kaggle. The goal is to understand these patterns to offer meaningful information to real estate market investors and participants. The study will delve into the various market dynamics and identify how housing prices vary according to characteristics like room counts, property dimensions, and other important features and criteria.

The project scope includes data analysis, cleaning, and manipulation to ensure accurate results. The project will use advanced regression techniques, such as Ridge Regression, Random Forest, ElasticNet, Lasso, and Support Vector Regression, to predict house prices based on different attributes. These predictive models will allow the user to make accurate assessments of house prices and provide buyers and sellers in the housing market with useful insight.

1.4 Aims and Objectives:

The primary goal of the project is to analyze housing market patterns and trends in Ames, Iowa. By employing advanced machine learning techniques, the project aims to uncover the factors that impact housing prices. It will identify key drivers for property values across various neighborhoods in Ames and assess different predictive models to find the most accurate one for forecasting future housing prices.

1.5 Research Questions:

During the study, the project will leverage data to understand the factors influencing housing prices. Some of the research questions that will be addressed include:

a) Which neighborhoods in Ames have the most expensive houses, and which have the most affordable ones?
b) What factors influence housing prices in Ames, such as the size of the property, number of bathrooms, or other attributes?

Chapter 2 - Literature Review

2.1 Literature Review:

The present section provides a discussion of key studies that focus on fitting appropriate models to the California Housing Data. The thesis contributes to a small yet growing collection of studies on machine learning in the housing market. There exists a diverse selection of literature pertaining to the prediction of housing prices, which includes both traditional regression models and autoregressive models. Only the machine learning literature that is directly related to this field is included [3, 4].

Accurately determining the value of a house is of utmost importance. The evaluation technique is vital for lending institutions in determining loan decisions. In order to mitigate financial losses resulting from loan defaults, these institutions must ensure that the assessed property values are commensurate with the loan amount. As part of the loan application process, lenders often need an inspection and appraisal. Multiple perspectives have been considered during the extensive examination of house appraisal.

The study conducted by Abigail Bola Adetunjia et al. (2022) delves into the realm of house price prediction using the Random Forest machine learning technique [5]. Due to its practical applicability in various real-world scenarios, the prediction of price variations rather than specific values has attracted considerable attention in the current real estate landscape. The authors assert that while the House Price Index (HPI) remains a common tool for estimating house price inconsistencies, it falls short in predicting individual housing prices accurately due to its reliance on repeat-sale transactions. This limitation necessitates the incorporation of additional information beyond HPI, considering factors such as location, city dynamics, and population demographics.

The methodology employed in this research involves utilizing the Random Forest algorithm to predict house prices, leveraging the UCI Machine Learning repository Boston housing dataset [5]. Through regression analysis and data exploration techniques, the study aims to forecast house prices based on relevant features provided in the dataset. Data pre-processing steps, including normalization and encoding, were undertaken to prepare the dataset for model development.

The findings show that, when compared to real values, the Random Forest model performs admirably in forecasting housing prices, with an acceptable error range of ±5 [5]. Performance evaluation metrics such as Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Root Mean Square Error (RMSE) validate the efficacy of the proposed model. The scatter plot visualization illustrates a satisfactory alignment between predicted and actual values, affirming the model's predictive prowess.

Furthermore, the study conducts a comprehensive exploration of various factors influencing house prices, including physical condition, location, and infrastructure development. Through data analysis and correlation matrix visualization, the research identifies key predictors of house prices, providing valuable insights for stakeholders in the real estate industry, policymakers, and urban planning authorities [5].

The research highlights the importance of machine learning methods, specifically Random Forest, in tackling the complexity involved in predicting property prices. By incorporating diverse features and leveraging advanced algorithms, the proposed model demonstrates the potential for enhancing decision-making processes in real estate investment and valuation. However, the authors acknowledge the need for further exploration of alternative machine learning models, including deep learning approaches, to enrich the predictive capabilities in the domain of house price forecasting [5].

Lu, S., Yang, (2017) explores the challenging task of predicting individual house prices using machine learning techniques and creative feature engineering [6]. Using data from the Kaggle Challenge "House Prices: Advanced Regression Techniques," the authors suggest a hybrid strategy that combines regression models from Lasso and Gradient Boosting. The goal is to improve prediction accuracy by considering various factors such as location, house type, size, build year, and local amenities.

The study starts out with an overview of machine learning methods and how they are used to forecast home values. It highlights the complexity of factors influencing house prices, including economic cycles, population movement, and interest rates. The authors emphasize the importance of data quality and feature engineering in accurately predicting house prices.

In the methodology section, the paper details the creative feature engineering process, which involves transforming numerical and categorical variables, handling missing values, and introducing new features based on domain knowledge. Various techniques such as log transformation and polynomial transformations are applied to enhance data normality and linearity.

The Ridge, Lasso, and Gradient Boosting regression methods were used in the study to do a thorough investigation of feature selection [6]. The ideal number of attributes, according to the authors, to obtain the highest score on the test data was 230. This conclusion was consistent across all three regression methods.

Ridge regression produced the best score on the test data with 230 features, whereas the minimal Root Mean Squared Error (RMSE) for the training data was noted with 160 features. In a similar vein, Lasso regression showed a similar pattern, with 160 features showing the lowest RMSE for the training data and 230 features showing the best score on the test data. The results from Gradient Boosting regression further supported this finding, with the optimal parameters for the training data suggesting a subsample of 0.5. However, a subsample of 0.6 resulted in a better score on the Kaggle test data, again indicating that 230 features yielded the most favourable outcome.

The discussion section emphasizes the importance of feature engineering in improving prediction accuracy. The study also suggests potential avenues for future research, such as incorporating additional variables like income, population demographics, and socioeconomic factors [6].

The study provides valuable insights into the application of machine learning techniques for house price prediction. By employing creative feature engineering and hybrid regression methods, the authors demonstrate significant improvements in prediction accuracy, underscoring the importance of thorough data preprocessing and model optimization.

In the study titled "Spatial Analysis of Housing Prices and Market Activity with the Geographically Weighted Regression" by Cellmer, Cichulska, and Bełej, published in the ISPRS International Journal of Geo-Information in 2020, the authors investigated the spatial dynamics of housing prices and market activity in Poland using Geographically Weighted Regression (GWR) [7]. They aimed to demonstrate that models incorporating spatial heterogeneity provide a better reflection of market relationships compared to conventional regression models.

The study utilized data from 380 counties in Poland (Local Administrative Units) for the year 2018. By employing GWR, the authors analysed the impact of socio-demographic, economic, and environmental factors on average housing property prices and the number of transactions in a spatial context. They compared the performance of GWR models with traditional Ordinary Least Squares (OLS) regression models, highlighting the importance of considering spatial variability in housing market determinants.

The findings of the study revealed that both average housing prices and market activity exhibited spatial autocorrelation, indicating spatial diversity in market dynamics across different regions of Poland. Through the application of GWR, the authors were able to identify specific local determinants of housing prices and market activity, emphasizing the spatially differentiated nature of these relationships.

Furthermore, the study highlighted the limitations of traditional global models in capturing spatial variations and underscored the effectiveness of GWR in providing valuable insights into local housing market dynamics [7]. The authors suggested that GWR models offer a more nuanced understanding of spatial heterogeneity and enable policymakers to make informed decisions tailored to specific regional contexts.

Overall, the research conducted by Cellmer, Cichulska, and Bełej contributes to advancing spatial analysis methodologies in the field of housing market research, particularly in understanding the complex interplay between socio-demographic, economic, and environmental factors at the local level. Their study underscores the importance of incorporating spatial considerations in housing market analysis and provides a framework for future research in this area.

2.2 Takeaway from Literature Reviews:

The literature reviews provided a wide range of studies focusing on predicting housing prices using various methods, particularly machine learning techniques, and regression models. Here are some key takeaways from these reviews:

- **Importance of Accurate Valuation:** Accurately determining the value of a house is crucial for lending institutions to make informed loan decisions, thus avoiding financial losses from defaulted loans.
- **Machine Learning in the Housing sector:** There is a rising amount of research focused on utilizing machine learning techniques to forecast housing values, highlighting its growing significance in the housing sector.
- **Model Comparison and Selection:** Studies conduct comparisons between various machine learning algorithms and regression models in order to determine the most efficacious ones for predicting house prices. These comparisons frequently take into account criteria such as model correctness, stability, and predictive capability.

- **Impact of Economic Variables:** House price forecasting is influenced by various economic factors, and accurate predictions have a direct impact on economic phenomena and systems.
- **Geographical Factors:** Geographical elements are crucial in determining property values. Research emphasizes their relevance, suggesting that spatial considerations significantly influence housing market dynamics.
- **Practical Implications:** The results of these research offer valuable guidance for sellers to set optimal prices for their homes and for purchasers to prevent paying excessively for residences.
- **Application of Machine Learning:** Machine learning is not only used for predicting housing market trends, but also finds applications in several other areas like medical diagnostics and stock market trading. This demonstrates the adaptability and efficiency of machine learning across different disciplines.

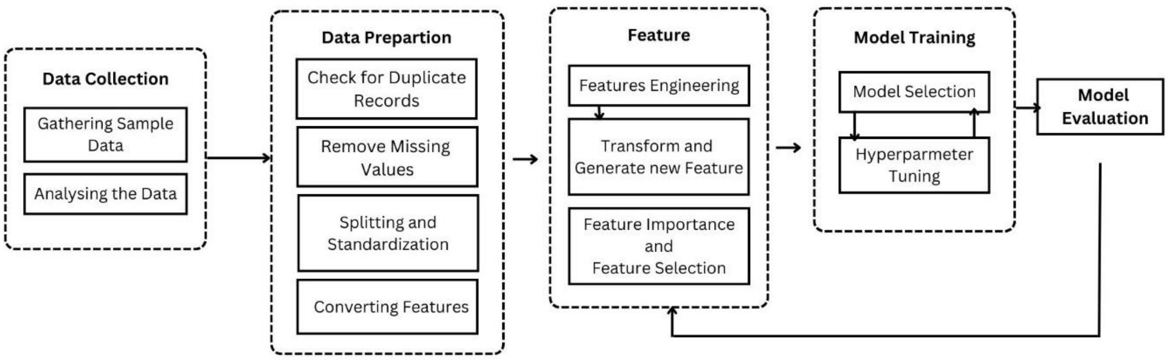
Overall, these literature reviews emphasize the importance of accurate prediction models in the housing market, the effectiveness of machine learning techniques, and the need for comprehensive analyses considering various factors affecting property values.

Chapter 3 – Research Methodology

3.1 Methodology:

The objective of this thesis is to utilize machine learning methodologies and competitive regression models to estimate the median house price within a district. Illustrated in Figure 3, the research framework for predicting house prices consists of five primary components: data collection, data preparation, feature processing, model training, and model evaluation. Each stage is elaborated upon in subsequent sections.

Fig. 3 - Research framework for the housing price



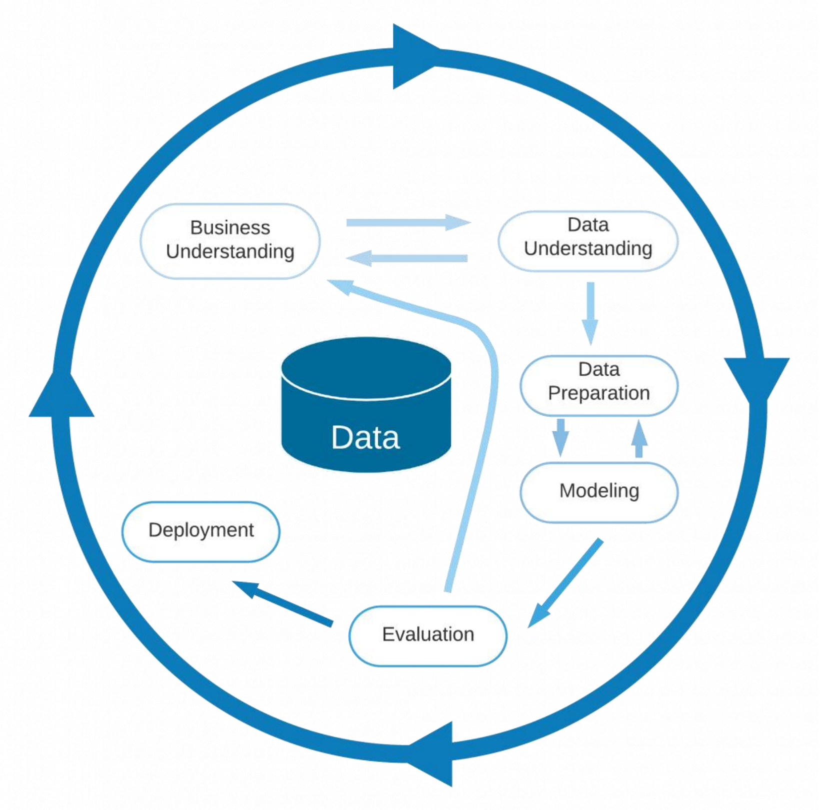
We're essentially trying out different tools to see which one works best for the job. We test various algorithms, such as Lasso, Ridge, Random Forest, and Support Vector Model to see which one can most accurately predict house prices. The aim is to gain a better understanding of how these methods function in the context of predicting real estate prices.

However, before we can get to that point, we need to clean and organize our data, similar to tidying up a messy room. Each house has unique characteristics that influence its price, so we carefully select which features are most important. It's like separating the essential items from the clutter.

There are lots of different methods we can use to solve various problems. These methods might work similarly well, but they can differ in how much computer power they need, how long they take to train, and the assumptions they make. So, the tricky part is figuring out which method is best for the problem you're trying to solve. Since real-world situations vary a lot in terms of the data available and what's needed, there's no one-size-fits-all solution.

To help researchers save time and pick the right methods, this thesis suggests a plan. It starts by introducing a common method called CRISP-DM, which stands for Cross-Industry Standard Process for Data Mining. This gives a general idea of how data mining and machine learning work. Then, it narrows down the focus to a specific task: predicting house prices. It lays out a step-by-step process for using supervised regression algorithms, which are the best choice for this kind of prediction.

Fig. 4 - Cross-Industry Standard Process for Data Mining (CRISP-DM)



https://www.kiunki.net/wp-content/uploads/2020/08/Data-Circle-1020x1024.png

Once the researcher identifies the problem, the subsequent step involves handling the data. This process encompasses more than just understanding the nature, functionality, and value of the data. Before analysis, data must be collected. Hence, upon clarifying the research goal, it becomes crucial to explore datasets relevant to the topic. While data quality is significant, the primary objective is to gain a comprehensive understanding of all potential data sources and features. After evaluating these options, it's vital to assess the dataset's suitability in achieving the research goal, either individually or collectively. Depending on the outcome of the data search, a project may necessitate additional data or may proceed with the available dataset.

3.2 Machine Learning Models:

Now talking about the methodologies we used in this program are Lasso Regression, Random Forest, Support Vector Regression, and Ridge Regression. Now we will discuss each of these methodologies.

3.2.1 Lasso Regression:

Lasso Regression, short for Least Absolute Shrinkage and Selection Operator Regression, is a regression technique that combines the concepts of linear regression with regularization to improve the model's performance and feature selection. It's particularly useful when dealing with datasets containing a large number of features, where feature selection and regularization are crucial for preventing overfitting and enhancing interpretability (Tibshirani, 1996) [8]. The primary objective of Lasso Regression is to minimize the sum of squared differences between the observed and predicted target variables, similar to traditional linear regression. However, Lasso Regression adds a regularization term to the objective function, penalizing the absolute values of the coefficients of the features (Hastie et al., 2009) [9].

Mathematical Formulation: The objective function of Lasso Regression can be expressed as: $\text{minimize} \left(\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right)$

where:

- yi is the observed target variable for the ith data point.
- y^i is the predicted target variable for the ith data point.
- βj is the coefficient of the jth feature.
- λ is the regularization parameter that controls the strength of the penalty term.

Regularization is a technique used to prevent overfitting by adding a penalty term to the model's loss function (Zou & Hastie, 2005) [10]. In Lasso Regression, the regularization term is the sum of the absolute values of the coefficients multiplied by a regularization parameter (λ or alpha). This penalty encourages the coefficients of less important features to shrink towards zero or exactly zero, effectively performing feature selection by eliminating irrelevant or redundant features. Because Lasso Regression penalizes the absolute values of the coefficients, it tends to force the coefficients of less important features to zero. As a result, Lasso Regression inherently performs feature selection, effectively identifying the most relevant features for predicting the target variable. Lasso Regression is a powerful technique for linear regression tasks, particularly when feature selection and regularization are important considerations. It strikes a balance between model complexity and interpretability, making it widely used in various fields, including machine learning, statistics, and economics.

3.2.2 Random Forest:

Random Forest Regression is a supervised learning technique that utilizes the ensemble learning approach to perform regression tasks (Segal, 2004) [12]. The ensemble learning method combines predictions generated by multiple machine learning algorithms to produce a forecast that is more accurate than a single model's prediction. During the training phase of a Random Forest operation, several decision trees are built (Breiman, 2001) [12]. 12Each decision tree is constructed using a random subset of the training data and a random subset of features, and the final prediction is obtained by averaging the predictions of all the individual trees. Mathematically, this can be represented as:

$$y\hat{} = N1\sum i = 1Nfi(x)$$

where $y\hat{}$ is the predicted target variable, N is the number of trees in the forest, and $fi(x)$ represents the prediction of the i-th decision tree. To improve prediction accuracy, the model aggregates the predictions of all trees, effectively reducing variance and overfitting. The Random Forest Regression model demonstrates robustness and accuracy in its analysis (Breiman, 2001). It is often capable of achieving outstanding outcomes in a broad range of scenarios, including ones that entail non-linear interactions. However, there are a few drawbacks, such as the lack of interpretability, the potential for overfitting, and the need to optimize parameters such as the number of trees in the model (Breiman, 2001).

3.2.3 Support Vector Regression:

Support Vector Regression (SVR) is a regression technique that extends the concepts of Support Vector Machines (SVMs) to the regression domain. Like SVMs for classification, SVR aims to find the optimal hyperplane that best fits the data while maximizing the margin [13]. Mathematically, SVR seeks to minimize the following objective function:

$$\min w, b, \xi, \xi * 21 \mid\mid w \mid\mid 2 + C\sum i = 1n(\xi i + \xi i *)$$

subject to the constraints:

- $yi-w\cdot\phi(xi)-b\leq\epsilon+\xi i$
- $w\cdot\phi(xi)+b-yi\leq\epsilon+\xi i*$
- $\xi i, \xi i * \geq 0$

where:

- w is the weight vector,
- b is the bias term,
- ξ and $\xi *$ are slack variables representing deviations from the margin,
- $\phi(x)$ is the feature mapping function,
- C is the regularization parameter, and
- ϵ is the epsilon-insensitive loss parameter.

In SVR, the margin is defined as a tolerance zone around the predicted values. The goal is to fit as many data points within this margin as possible, while minimizing the deviation of data points outside the margin. The data points that lie on the boundary of the margin are called support vectors, as they influence the position and orientation of the regression hyperplane. SVR aims to minimize the error between the predicted and actual target values, subject to the margin constraints. This is achieved by minimizing a loss function, which penalizes deviations from the target values within the margin. Overall, Support Vector Regression is a powerful technique for regression tasks, particularly when dealing with complex relationships and high-dimensional data. By optimizing a margin of tolerance around the predicted values, SVR aims to find a robust regression function that generalizes well to unseen data [14].

3.2.4 Ridge Regression:

Ridge Regression is a linear regression technique that addresses the issue of multicollinearity by introducing a regularization term to the least squares objective function (Hoerl & Kennard, 1970) [15]. The primary objective of Ridge Regression is to minimize the sum of squared differences between the observed and predicted target variables, similar to traditional linear regression. However, Ridge Regression adds a penalty term to the objective function, which is proportional to the square of the magnitude of the coefficients. Mathematically, the objective function of Ridge Regression can be expressed as:

$$min\beta(\sum i = 1n(yi - Xi\beta)2 + \lambda\sum j = 1p\beta j2)$$

where:

- yi is the observed target variable for the i-th data point,
- Xi is the feature vector for the i-th data point,
- β is the coefficient vector,
- λ is the regularization parameter that controls the strength of the penalty term, and
- p is the number of features.

The penalty term shrinks the coefficients towards zero, reducing their variance and helping to alleviate multicollinearity issues (Hastie et al., 2009) [16]. As a result, Ridge Regression produces more stable estimates, particularly when dealing with high-dimensional datasets or datasets with highly correlated features. Ridge Regression strikes a balance between bias and variance, making it a useful tool for regression tasks, especially in scenarios where multicollinearity is a concern.

Chapter 4 – Data Analysis

4.1 Data Overview:

The dataset that I have used in this project is the "House Prices — Advanced Regression Techniques" dataset. It is one of the most popular datasets on Kaggle. It consists of two main datasets: "train" and "test." The primary objective is to predict the sale prices of residential properties based on a wide array of physical attributes and characteristics. The training dataset contains 1460 observations, while the test dataset contains 1459 observations. The dataset was originally compiled by Dean De Cock for a regression competition hosted on Kaggle. It was created to provide a real-world dataset for practitioners to apply regression techniques and predict housing prices. [17]

CSV Files Descriptions:

- **train.csv**: The training dataset comprises a total of 1460 observations, with each row representing a unique residential property. This dataset includes features describing various aspects of the properties alongside the target variable, SalePrice.
- **test.csv**: The test dataset contains 1459 observations, with the same set of features as the training dataset but without the target variable. The goal is to predict the sale prices of these properties using trained models.

Data Fields Descriptions:

The dataset consists of a comprehensive set of features, including both numerical and categorical variables. Here are the key features:

1. **SalePrice**: The sale price of the property in dollars (target variable).
2. **MSSubClass**: The building class.
3. **MSZoning**: The general zoning classification.
4. **LotFrontage**: Linear feet of street connected to the property.

5. **LotArea**: Lot size in square feet.
6. **Street**: Type of road access.
7. **Alley**: Type of alley access.
8. **LotShape**: General shape of the property.
9. **LandContour**: Flatness of the property.
10. **Utilities**: Type of utilities available.
11. **LotConfig**: Lot configuration.
12. **LandSlope**: Slope of the property.
13. **Neighborhood**: Physical locations within Ames city limits.
14. **Condition1** and **Condition2**: Proximity to main roads or railroad.
15. **BldgType**: Type of dwelling.
16. **HouseStyle**: Style of dwelling.
17. **OverallQual**: Overall material and finish quality.
18. **OverallCond**: Overall condition rating.
19. **YearBuilt**: Original construction date.
20. **YearRemodAdd**: Remodel date.
21. **RoofStyle**: Type of roof.
22. **RoofMatl**: Roof material.
23. **Exterior1st** and **Exterior2nd**: Exterior covering on the house.
24. **MasVnrType** and **MasVnrArea**: Masonry veneer type and area.
25. **ExterQual** and **ExterCond**: Exterior material quality and condition.
26. **Foundation**: Type of foundation.
27. **BsmtQual**, **BsmtCond**, **BsmtExposure**, **BsmtFinType1**, **BsmtFinSF1**, **BsmtFinType2**, **BsmtFinSF2**, **BsmtUnfSF**, and **TotalBsmtSF**: Basement-related features.

28. **Heating**, **HeatingQC**, **CentralAir**: Heating and air conditioning features.
29. **Electrical**: Electrical system.
30. **1stFlrSF**, **2ndFlrSF**, **LowQualFinSF**, and **GrLivArea**: Living area square footage.
31. **BsmtFullBath**, **BsmtHalfBath**, **FullBath**, and **HalfBath**: Bathroom features.
32. **BedroomAbvGr**: Number of bedrooms above basement level.
33. **KitchenAbvGr**, **KitchenQual**: Kitchen-related features.
34. **TotRmsAbvGrd**: Total rooms above grade (excluding bathrooms).
35. **Functional**: Home functionality rating.
36. **Fireplaces**: Number of fireplaces.
37. **FireplaceQu**: Fireplace quality.
38. **GarageType**, **GarageYrBlt**, **GarageFinish**, **GarageCars**, **GarageArea**, **GarageQual**, and **GarageCond**: Garage-related features.
39. **PavedDrive**: Paved driveway.
40. **WoodDeckSF**, **OpenPorchSF**, **EnclosedPorch**, **3SsnPorch**, **ScreenPorch**, and **PoolArea**: Outdoor features.
41. **PoolQC**, **Fence**, **MiscFeature**, and **MiscVal**: Miscellaneous features and their values.
42. **MoSold** and **YrSold**: Month and year of sale.
43. **SaleType** and **SaleCondition**: Type of sale and sale condition.

Overall, this dataset provides a comprehensive view of residential properties attributes, and best dataset for detailed analysis and predictive modeling to understand and forecast housing prices accurately.

4.2 Exploratory Data Analysis (EDA)

EDA stands for Exploratory Data Analysis. It's a process where we examine a dataset to understand its main characteristics, often before applying more formal statistical techniques. EDA involves looking at the data in various ways, such as plotting graphs, calculating summary statistics, and identifying patterns or trends. The goal of EDA is to gain insights into the data that can guide further analysis or decision-making. It helps us understand what our data looks like and what kind of information it contains.

The EDA conducted on the "House Prices — Advanced Regression Techniques" dataset provides a comprehensive understanding of the dataset's characteristics, distributions, relationships between features, and potential patterns or trends. I used statistical and graphical representations of the data to understand the relationship between features and prices.

Firstly, I have checked the shape of the datasets, which provide the detail about the number of rows and columns in the dataset.

The output of the .shape() is :

The training dataset contains 1460 rows and 81 columns.
The test dataset contains 1459 rows and 80 columns.

Then I printed the first 5 rows of the training and test dataset using .head()

This enables us to print the names of all the columns in the dataset's first 5 rows, which helps us understand what each variable represents in the dataset throughout the program.

For train dataset:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2008	WD	Normal	208500
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	5	2007	WD	Normal	181500
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	9	2008	WD	Normal	223500
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2006	WD	Abnorml	140000
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	12	2008	WD	Normal	250000

5 rows × 81 columns

Fig. 5 - Train Dataset Fields

For test dataset:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	ScreenPorch	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	YrSold	SaleType	SaleCondition
0	1461	20	RH	80.0	11622	Pave	NaN	Reg	Lvl	AllPub	...	120	0	NaN	MnPrv	NaN	0	6	2010	WD	Normal
1	1462	20	RL	81.0	14267	Pave	NaN	IR1	Lvl	AllPub	...	0	0	NaN	NaN	Gar2	12500	6	2010	WD	Normal
2	1463	60	RL	74.0	13830	Pave	NaN	IR1	Lvl	AllPub	...	0	0	NaN	MnPrv	NaN	0	3	2010	WD	Normal
3	1464	60	RL	78.0	9978	Pave	NaN	IR1	Lvl	AllPub	...	0	0	NaN	NaN	NaN	0	6	2010	WD	Normal
4	1465	120	RL	43.0	5005	Pave	NaN	IR1	HLS	AllPub	...	144	0	NaN	NaN	NaN	0	1	2010	WD	Normal

5 rows × 80 columns

Fig.6 – Test Dataset Fields

Only the SalePrice column is missing in the test dataset.

In the next step, it showed the Range of the Data entries, which is 0 to 1459, and the number of data columns, which is 81. Also, the datatype of each column as we can see in Fig 7.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 81 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             1460 non-null   int64
 1   MSSubClass     1460 non-null   int64
 2   MSZoning       1460 non-null   object
 3   LotFrontage    1201 non-null   float64
 4   LotArea        1460 non-null   int64
 5   Street         1460 non-null   object
 6   Alley          91 non-null     object
 7   LotShape       1460 non-null   object
 8   LandContour    1460 non-null   object
 9   Utilities      1460 non-null   object
 10  LotConfig      1460 non-null   object
 11  LandSlope      1460 non-null   object
 12  Neighborhood   1460 non-null   object
 13  Condition1     1460 non-null   object
 14  Condition2     1460 non-null   object
 15  BldgType       1460 non-null   object
 16  HouseStyle     1460 non-null   object
 17  OverallQual    1460 non-null   int64
 18  OverallCond    1460 non-null   int64
 19  YearBuilt      1460 non-null   int64
 20  YearRemodAdd   1460 non-null   int64
 21  RoofStyle      1460 non-null   object
 22  RoofMatl       1460 non-null   object
 23  Exterior1st    1460 non-null   object
 24  Exterior2nd    1460 non-null   object
 25  MasVnrType     1452 non-null   object
 26  MasVnrArea     1452 non-null   float64
 27  ExterQual      1460 non-null   object
 28  ExterCond      1460 non-null   object
 29  Foundation     1460 non-null   object
 30  BsmtQual       1423 non-null   object
 31  BsmtCond       1423 non-null   object
 32  BsmtExposure   1422 non-null   object
 33  BsmtFinType1   1423 non-null   object
 34  BsmtFinSF1     1460 non-null   int64
 35  BsmtFinType2   1422 non-null   object
 36  BsmtFinSF2     1460 non-null   int64
 37  BsmtUnfSF      1460 non-null   int64
 38  TotalBsmtSF    1460 non-null   int64
 39  Heating        1460 non-null   object
 40  HeatingQC      1460 non-null   object
 41  CentralAir     1460 non-null   object
 42  Electrical     1459 non-null   object
 43  1stFlrSF       1460 non-null   int64
 44  2ndFlrSF       1460 non-null   int64
 45  LowQualFinSF   1460 non-null   int64
 46  GrLivArea      1460 non-null   int64
 47  BsmtFullBath   1460 non-null   int64
 48  BsmtHalfBath   1460 non-null   int64
 49  FullBath       1460 non-null   int64
 50  HalfBath       1460 non-null   int64
 51  BedroomAbvGr   1460 non-null   int64
 52  KitchenAbvGr   1460 non-null   int64
 53  KitchenQual    1460 non-null   object
 54  TotRmsAbvGrd   1460 non-null   int64
 55  Functional     1460 non-null   object
 56  Fireplaces     1460 non-null   int64
 57  FireplaceQu    770 non-null    object
 58  GarageType     1379 non-null   object
 59  GarageYrBlt    1379 non-null   float64
 60  GarageFinish   1379 non-null   object
 61  GarageCars     1460 non-null   int64
 62  GarageArea     1460 non-null   int64
 63  GarageQual     1379 non-null   object
 64  GarageCond     1379 non-null   object
 65  PavedDrive     1460 non-null   object
 66  WoodDeckSF     1460 non-null   int64
 67  OpenPorchSF    1460 non-null   int64
 68  EnclosedPorch  1460 non-null   int64
 69  3SsnPorch      1460 non-null   int64
 70  ScreenPorch    1460 non-null   int64
 71  PoolArea       1460 non-null   int64
 72  PoolQC         7 non-null      object
 73  Fence          281 non-null    object
 74  MiscFeature    54 non-null     object
 75  MiscVal        1460 non-null   int64
 76  MoSold         1460 non-null   int64
 77  YrSold         1460 non-null   int64
 78  SaleType       1460 non-null   object
 79  SaleCondition  1460 non-null   object
 80  SalePrice      1460 non-null   int64
dtypes: float64(3), int64(35), object(43)
memory usage: 3.9 MB
None
```

Fig 7. Datatype of columns

We can see that I dataset has 3 float values, 35 int values, and 43 object values.

Moving Forward to the graphical representations of data, the analysis focuses on establishing relationships between SalePrice and other features. This includes scatter plots, correlation matrices, and histograms, elucidating how various factors influence house prices.

Firstly, Let’s see how expensive are houses in this Dataset. Fig. 8 Shows the Sale Price of the Houses and the frequency, like how many houses are in a particular price range. This will help buyers, sellers, and property agents to understand the price range in Ames Lowa.

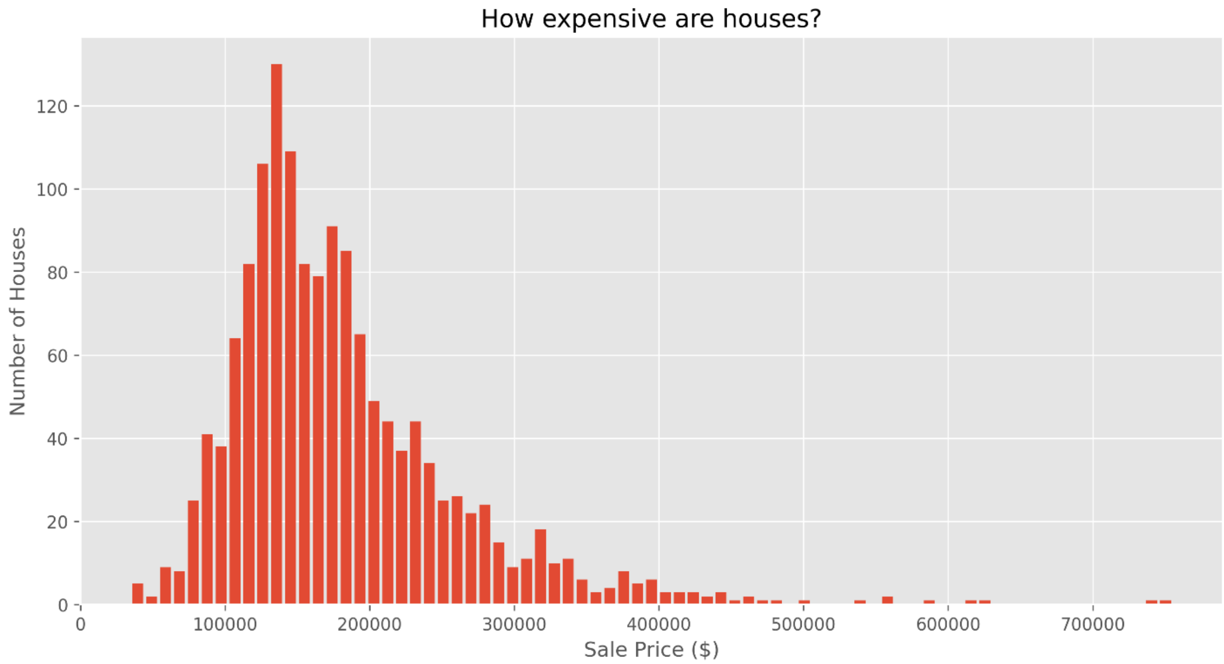


Fig. 8 – Sale Price of the houses

Looking at the prices, it's clear that most houses fall in the range of $100,000 to $200,000. I also found the cheapest house sold for $34,900 and the most expensive one for $755,000. On average, houses were sold for around $180,921, but if we look at the middle price, which is the median, it's $163,000.

Next, let’s figure out how old the houses are. In the Fig. 9, we will find the age of the houses. Where I tried to show the Distribution of the houses according to the year they were built.

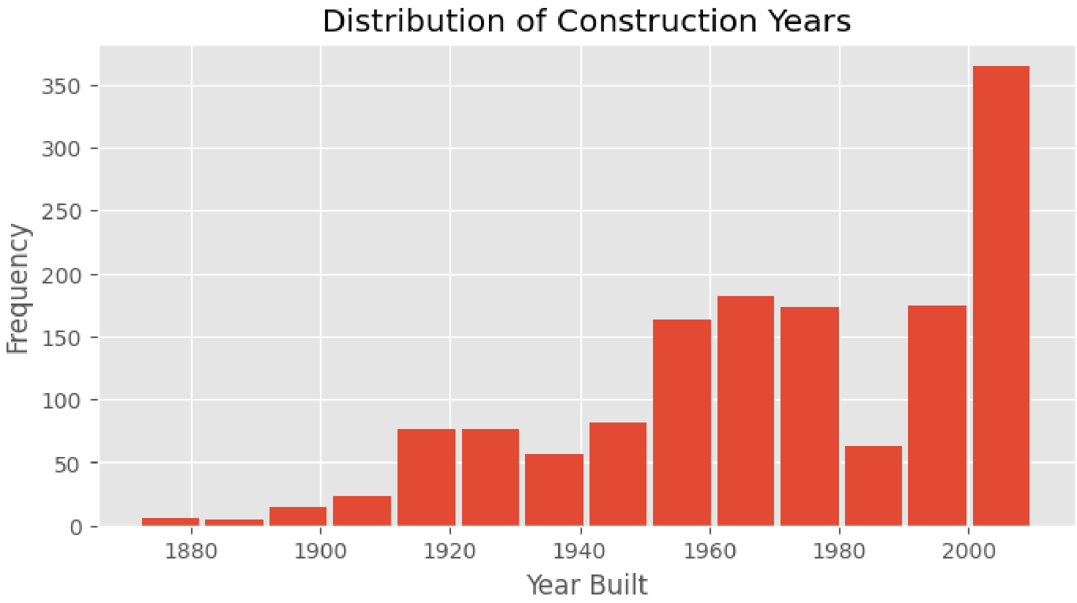


Fig. 9 - Distribution of Construction Years

The dataset reveals a fascinating trend in housing construction spanning from the 1880s to 2010 in Ames, Iowa. Notably, the majority of homes were built between the 1950s and 2010, indicating significant growth and development during this period. The oldest house in the dataset dates back to 1872, highlighting the longstanding history of residential architecture in the region, while the most recent construction occurred in 2010.

One thing that stands out is that in 2010, a lot of new houses were built—more than 350. This suggests that there was a big need for houses around that time. Maybe more people wanted to move to Ames, or there were more families needing homes. It could be because of things like how the economy was doing or changes in the population.

Examining the historical data, it's evident that the 1880s saw a relatively low rate of new construction, which could be attributed to various factors such as limited population growth or subdued demand. However, the 1920s witnessed a remarkable boom in the housing market, characterized by frequent and substantial construction activity.

Now moving forward, if we can check the Sales trend according to months then we can find a better understanding of sales patterns during the year. Fig. 10 shows the insight of sales trends during 2006 to 2010 each month.

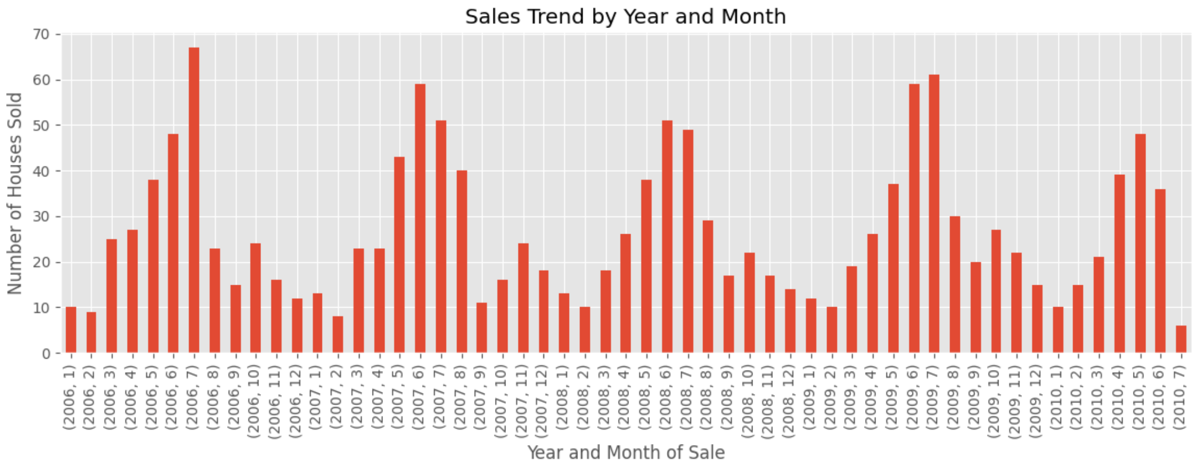


Fig. 10 – Sales trend by year and month

It's fascinating to note that a clear seasonal trend in house sales, with noticeable peaks occurring in May, June, and July each year. We've confirmed that the dataset covers the period from 2006 to 2010. I want to mention that we have only data till July 2010.

We can see that each year starts with a low Sales trend and during the time it’s increased and mid-year, it’s on peak and then we can see a sudden downfall in the graph. It’s mean most people prefer to buy a house during summer.

It's an interesting insight that highlights the preference for buying houses during summer season.

So far, we've explored the cost and construction years of houses, as well as the timing of their sales each year. Now, let's delve into another aspect: the size of these houses.

In Fig. 11, we're going to examine the data on house sizes. This will give us insight into whether there's more demand for larger or smaller houses. This graph shows the distribution of house sizes and how often each size occurs.

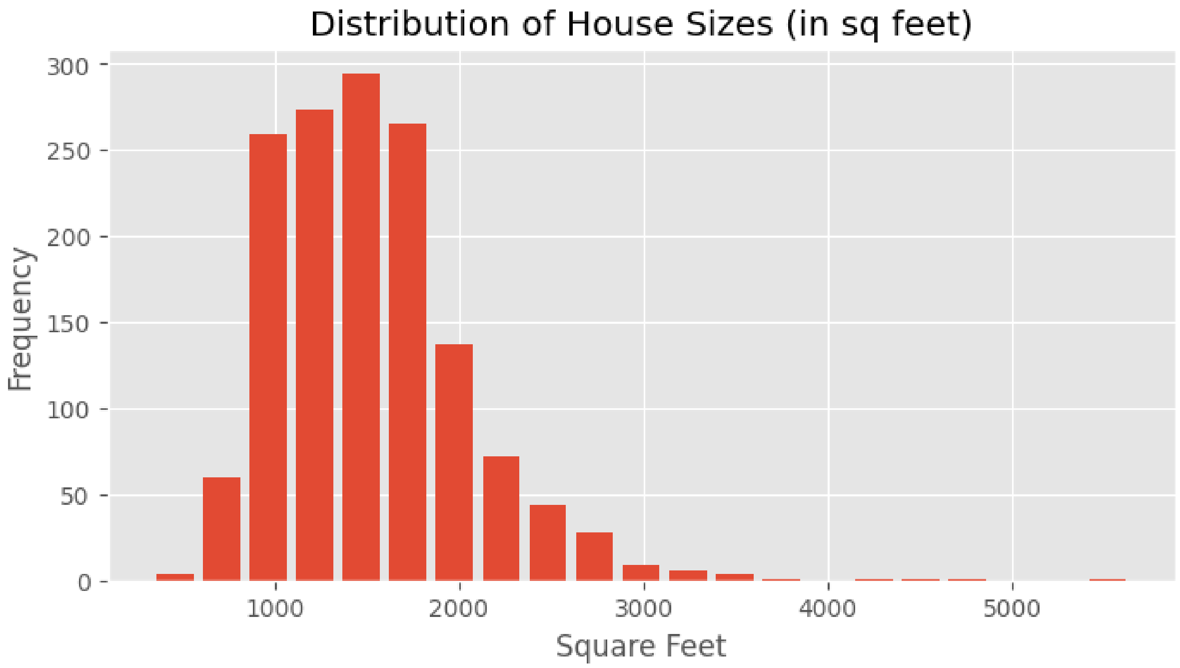


Fig. 11 – Distribution of house sizes

Looking at the sizes of the houses, we can see that there are nearly 300 houses with 1800 square feet, followed by around 260 houses with 1600 square feet. The largest house has 5,642 square feet of space, while the smallest has only 334 square feet. On average, the houses have an area of 1,515 square feet, while the middle value, or median, is slightly smaller at 1,464 square feet.

Before moving on to Data Processing, let’s do some more Analysis such as Univariate Analysis. According to Agresti, A., & Finlay, B. (2018),[18] Univariate analysis refers to the analysis of a single variable or attribute at a time. It's a fundamental technique used in statistics to understand the characteristics and properties of a single variable without considering its relationship with other variables.

In the context of our dataset, let’s do a univariate analysis on SalePrice. Conducting a univariate analysis of the SalePrice variable would involve examining various statistical measures and visualizations to understand its distribution. I choose skewness, and kurtosis to describe the central tendency, spread, and shape of the SalePrice distribution.

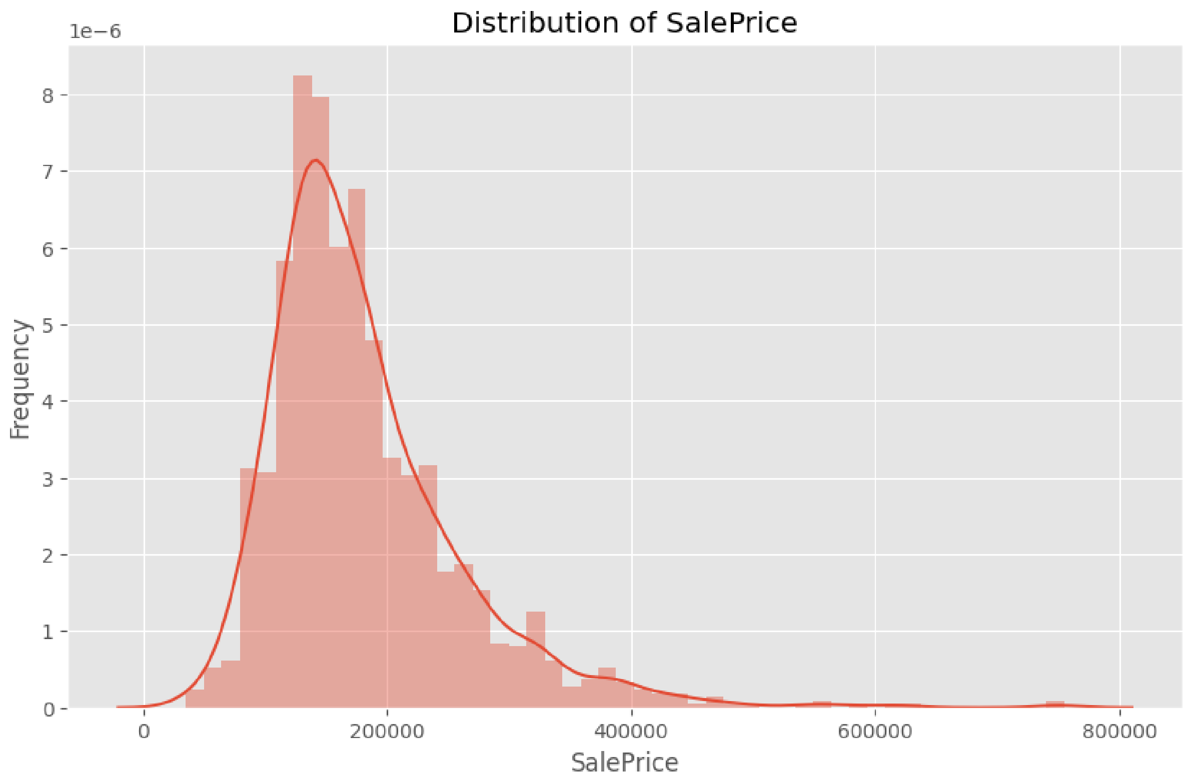


Fig. 12 – SalePrice Distribution

As we can see in Fig. 12, the SalePrice distribution exhibits positive skewness with a skewness value of 1.883, indicating that the distribution is skewed to the right. Additionally, the kurtosis value of 6.536 suggests that the distribution is “peaky” with fat tails, implying the presence of outliers, particularly on the right side of the distribution.

This means that the majority of house prices are clustered towards the lower end of the scale, with a few exceptionally high-priced outliers pulling the mean towards the right.

Like this, I tried to show the distribution of all the features, and I noticed that several features in our dataset that seem to have a positive skew, much like SalePrice. These include: LotFrontage, LotArea, BsmtUnfSF, TotalBsmtSF, 1stFlrSF, GrLivArea, and GarageArea as shown in Fig. 13.

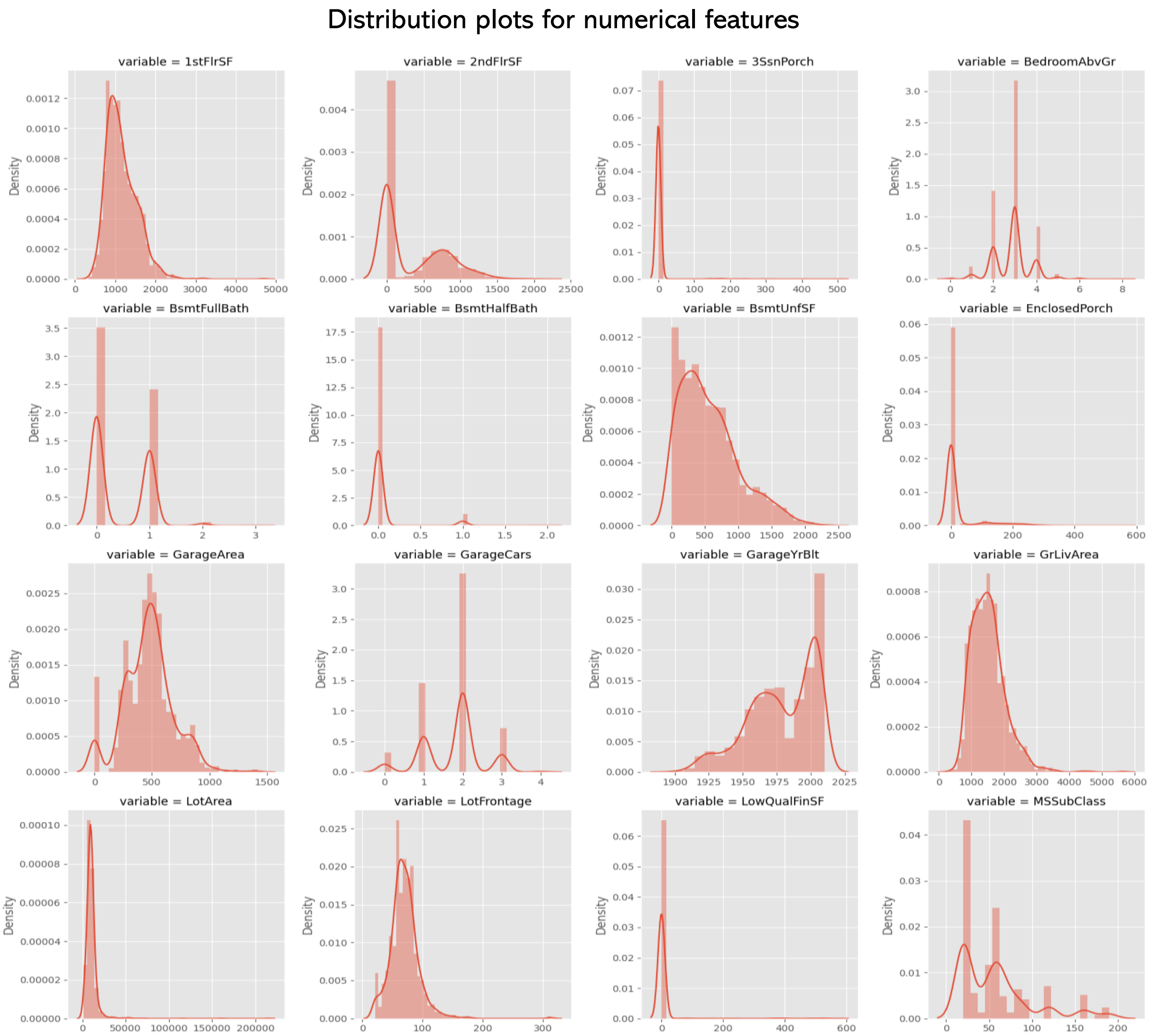


Fig. 13 - Distribution plots for numerical features

That's all about Exploratory Data Analysis. Now we can move on to the Data Processing, where we will see how we handle the missing values, categorical features to num features.

4.3 Data Preprocessing:

Data preprocessing is a crucial step in the data analysis pipeline, involving various techniques to clean, transform, and prepare raw data for analysis and modeling. The process begins with data cleaning, which includes handling missing values, outliers, duplicates, and other errors to ensure data integrity. [19]

Once cleaned, the data may require transformation to meet the assumptions of statistical techniques or improve the performance of machine learning models. This often involves normalization, log transformation, and encoding categorical variables. Feature engineering may then be applied to create new features or modify existing ones to better capture underlying patterns. [20]

Dimensionality reduction techniques, such as PCA, can help reduce the number of features while preserving important information. Additionally, data integration may be necessary to combine data from multiple sources, while data normalization ensures consistency in scale across numerical features [21]. Overall, data preprocessing is essential for producing clean, consistent, and relevant data that facilitates accurate analysis and decision-making.

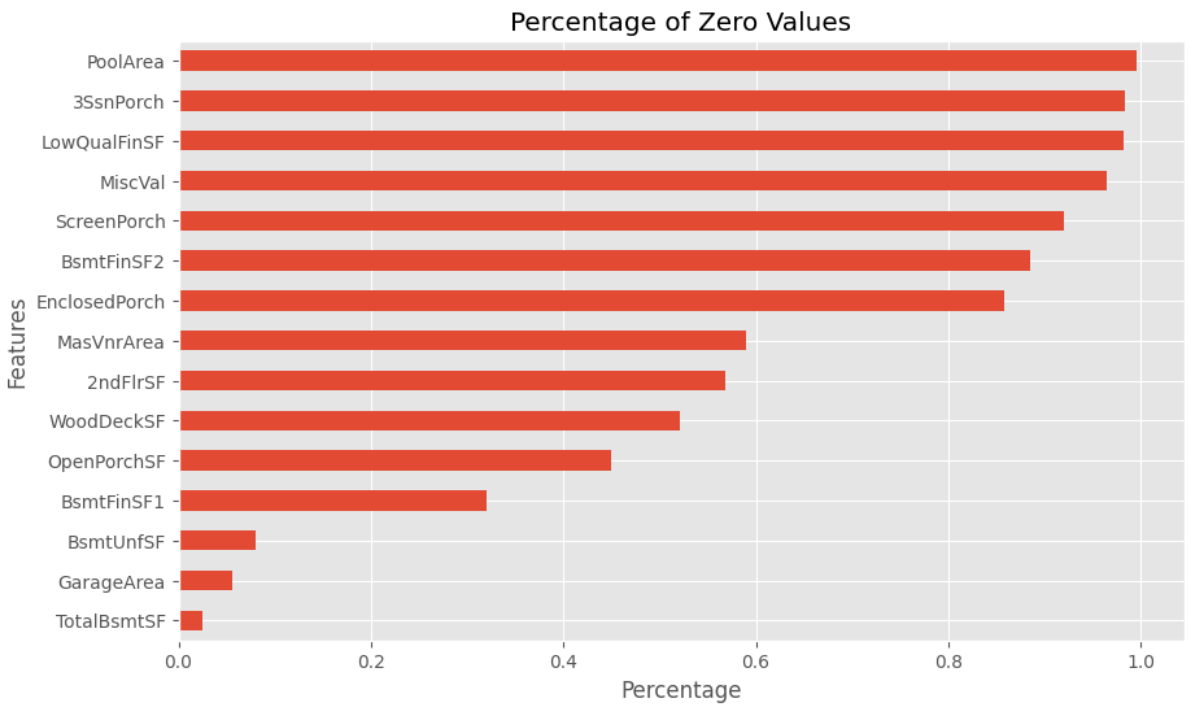


Fig. 14 – Percentage of Zero Values

In Figure 14, we observe that many features have a substantial number of zero values, indicating that numerous houses lack these features altogether. For instance, attributes such as Pool, Miscellaneous feature, Screen Porch, Wood Deck, 3 Season Porch, and Garage Area show a significant portion of houses with zero values. Notably, over 95% of houses in Ames, Lowa, are devoid of a pool area, highlighting the rarity of this feature among residential properties. Similarly, the absence of the Garage Area is notable, with approximately 7-8% of houses in Ames, Lowa, lacking this facility.

4.3.1 Feature Engineering

Fig. 15 – Categorical Features

Fig. 15 shows the categorical features of the houses, which provide descriptive information about the quality of various amenities. For instance, it indicates whether a house has an air conditioning system, represented as "Y" for yes and "N" for no. Similarly, it showcases the heating quality rating, ranging from excellent to poor. The graph presents a plethora of features and their corresponding quality ratings and categorical attributes.

But to enhance prediction accuracy, converting categorical features into numerical ones is crucial. I achieved this through custom data transformation by mapping each categorical feature's values to alternative numerical values. For ordinal features such as heating quality ratings (e.g., excellent, good, fair, poor), I assigned numerical values corresponding to their qualitative order, ensuring that higher ratings received higher numerical values. You can see this in Fig. 16, where I tried to show some of converted num features. This transformation enables machine learning algorithms to process and interpret categorical information effectively, ultimately improving the predictive power of the model.

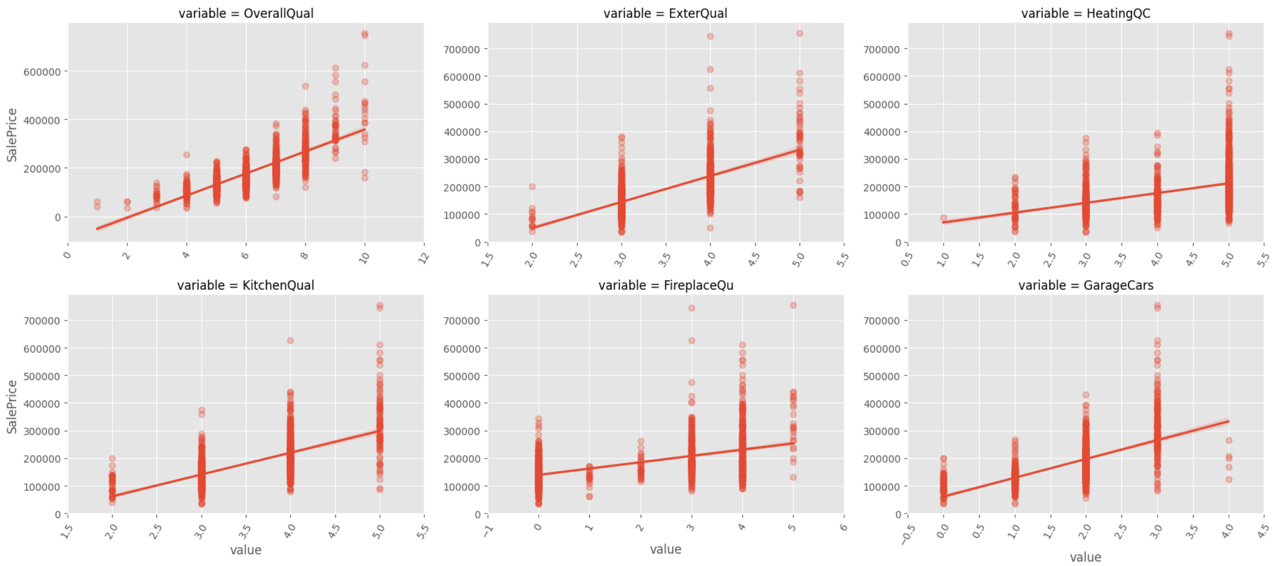


Fig. 16 – Num Features

These values represent the quality of various features with the sale price of houses. A notable trend emerges, showcasing that as the quality of features increases, so does the corresponding sale price. For instance, higher-rated features such as Overall Quality, Exterior Quality, Kitchen Quality, Basement Quality, and Garage Finish tend to coincide with higher sale prices. This correlation underscores the significance of feature quality in determining the value and desirability of houses in the market. Also, we can see this in Correlation Heatmap in Fig. 17.

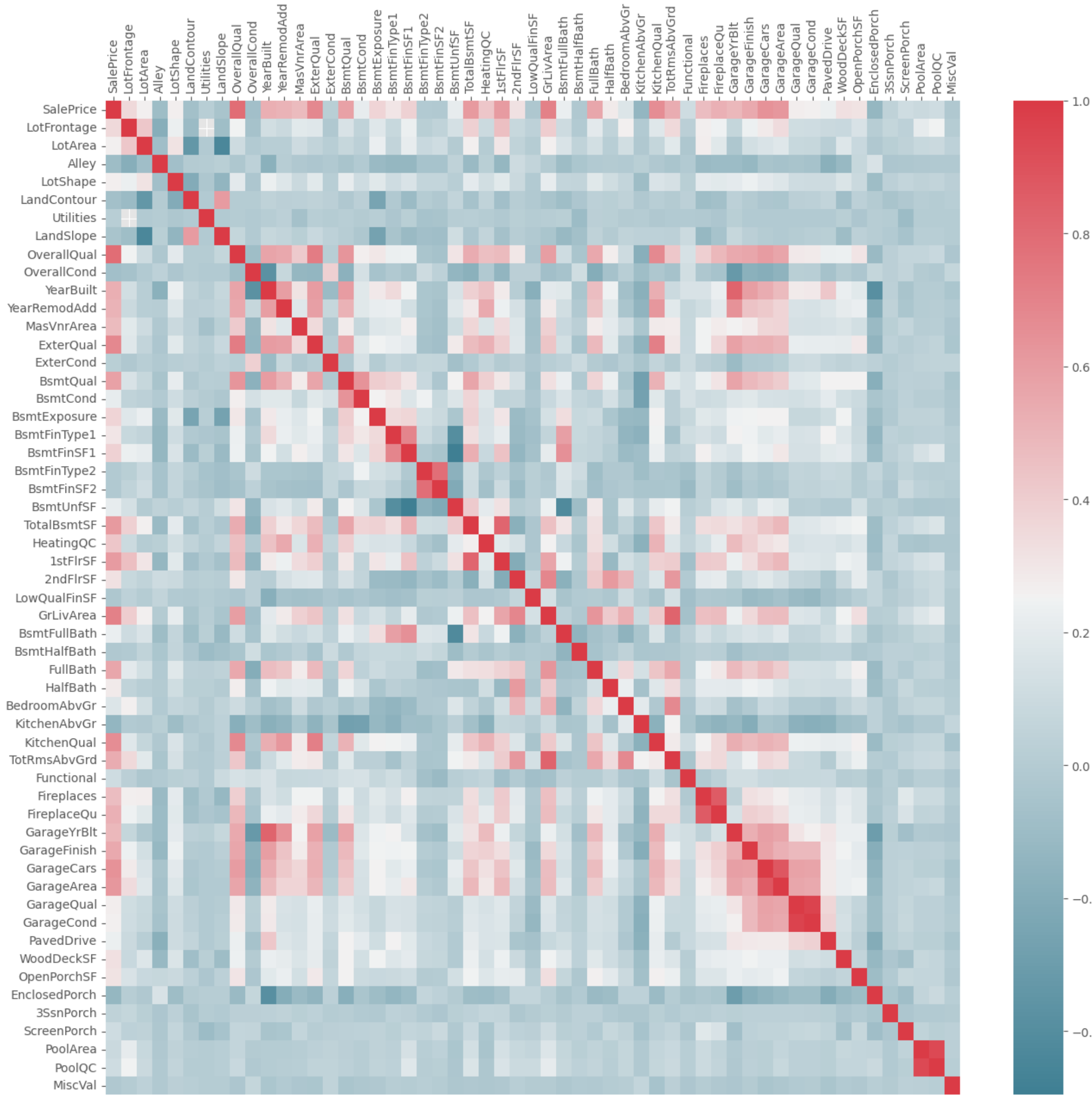


Fig. 17 – Correlation Heatmap of Features

I observe significant correlations between various features, including both expected and unexpected relationships. For instance, the high correlation between GarageYrBlt and YearBuilt indicates that garages are typically constructed concurrently with the main house. Similarly, the correlation between BsmtQual and OverallQual, as well as TotalBsmtSF and 1stFlrSF, aligns with intuitive expectations. Additionally, the strong correlation between KitchenQual and ExteriorQual, while seemingly unrelated, may suggest a hidden variable such as "new house," wherein newer homes tend to exhibit superior qualities in both kitchen and exterior features.

Now, let's delve deeper into how these features correlate with SalePrice, providing insights into which aspects of a property most strongly influence its market value. Fig. 18.

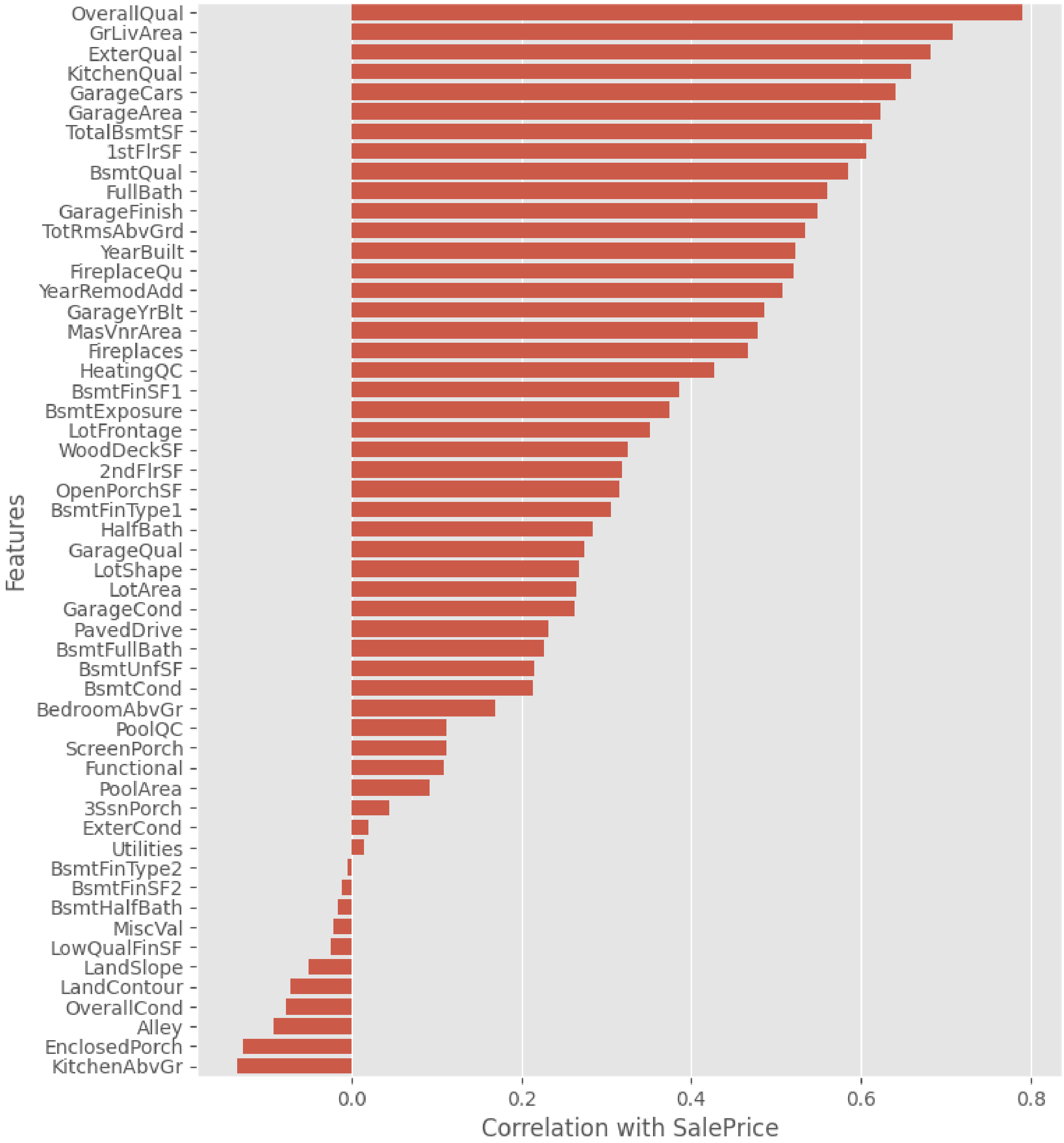


Fig. 18 – Features Correlation with Sale Price

The feature that shows the strongest correlation with the sale price is OverallQual, which is understandable. After this, the size of the house (GrLivArea) comes next then, there are three more features related to quality: ExterQual, KitchenQual, and BsmtQual. I mentioned earlier that ExterQual and KitchenQual are highly related, and now we see that they are also strongly linked to the sale price.

One last plot that correlates Sale price with categorical features. For this, I created a box plot because it effectively shows the distribution of sale prices across different categories of a categorical variable. Fig. 19.

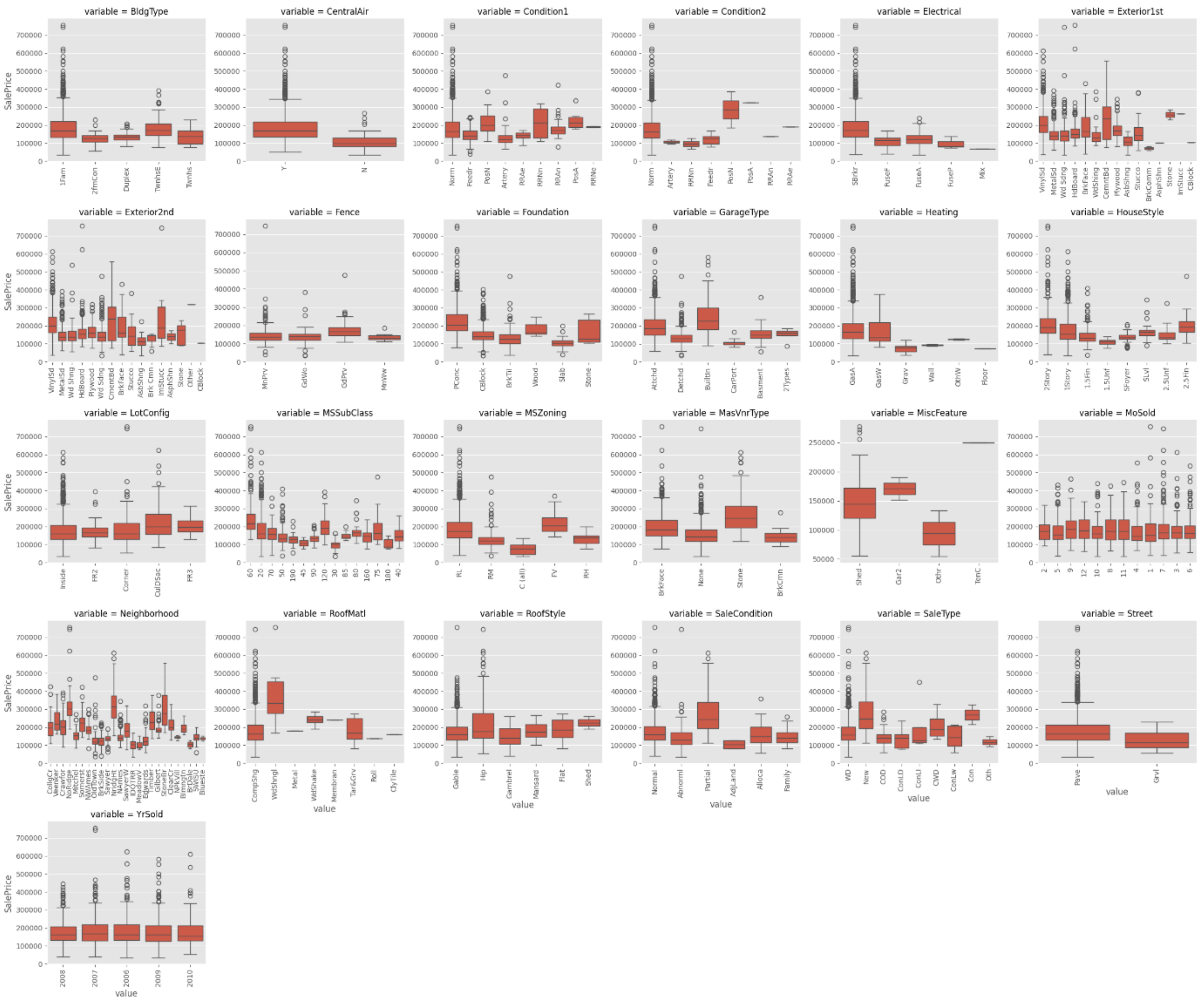


Fig. 19 – Correlation between Categorical features and Sales Price

It looks like some features show significant variance in the mean of SalePrice between different groups, eg. Neighborhood, SaleType or MSSubClass. We're aiming to determine which categorical features have a significant impact on SalePrice. To do this, we'll conduct one-way ANOVA tests for each feature against SalePrice. This test provides both F-statistics and p-values. A higher F-statistic indicates a stronger relationship, while a lower p-value suggests higher confidence in rejecting the null hypothesis. Essentially, the p-value tells us how likely it is to observe data if each group had no effect on SalePrice. The lower the p-value, the more significant the feature's influence on SalePrice. We'll sort the features based on their p-values to identify the most influential ones.

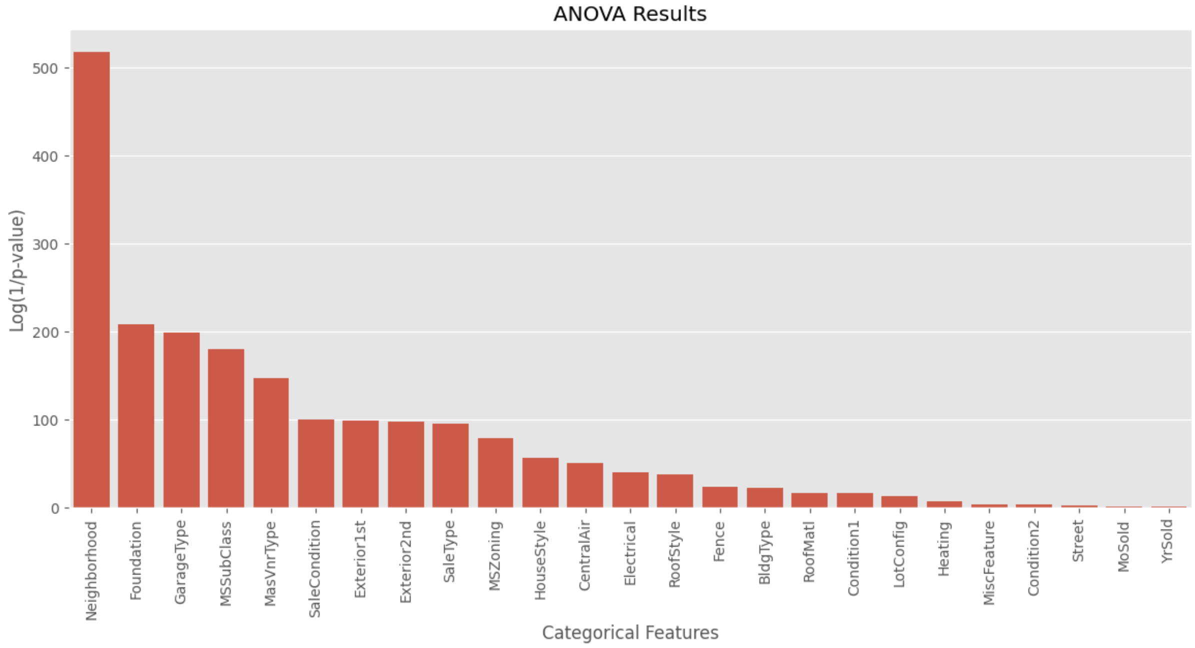


Fig. 20 – ANOVA Results

Location is the most crucial thing to determine the house price. Among all our categorical features, Neighborhood stands out as having the most significant impact on SalePrice. We can see this in Fig. 20.

It's crucial to emphasize the overwhelming influence of the Neighborhood. By taking the logarithm of the inverse of the p-value (np.log(1./anova['p'])), we ensure positive values and a more informative visualization. In essence, the p-value for Neighborhood is approximately 300 times smaller than that of the next feature, underscoring its unparalleled significance.

4.3.2 Handling Missing Values:

Throughout our analysis, we've addressed missing values in several stages. Initially, when converting certain categorical features to numerical representations, we filled missing values with 0. This decision was logical as these categorical features were rank-based, and assigning 0 to missing values maintained the ordinality of the data. This approach was applied to the following features: Alley, BsmtCond, BsmtExposure, BsmtFinType1, BsmtFinType2, BsmtQual, ExterCond, ExterQual, FireplaceQu, Functional, GarageCond, GarageFinish, GarageQual, HeatingQC, KitchenQual, LandContour, LandSlope, LotShape, PavedDrive, PoolQC, and Utilities.

Subsequently, before conducting the ANOVA evaluation of categorical features, we replaced missing values with "Missing". This step ensured that missing values were treated as a separate category, allowing us to evaluate their impact on SalePrice independently.

I have checked the categorical features with "Missing" values and I got;

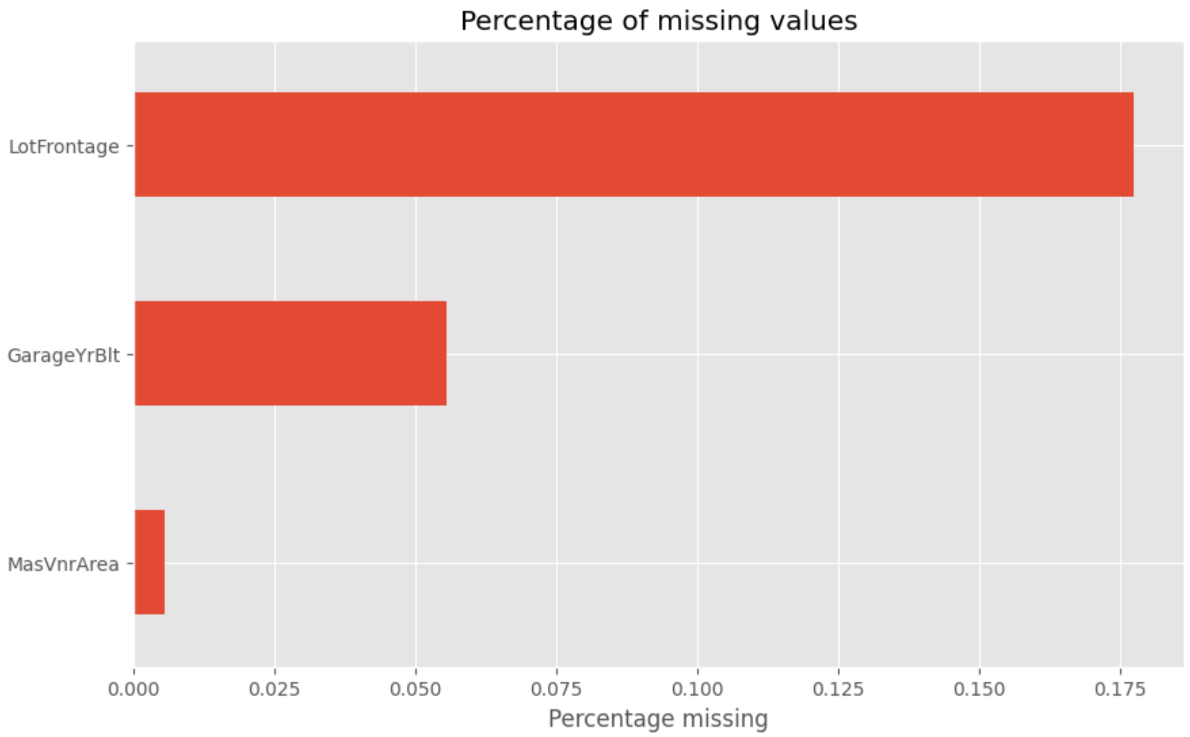


Fig. 21 – Percentage of missing values

We can see that, LotFrontage has the most missing values, followed by the GarageYrBlt and MasVnrArea. I filled the missing values in LotFrontage with the median LotFrontage of the corresponding neighborhood. I transformed each group by filling missing values with the median value of 'LotFrontage' within that group. The lambda x: x.fillna(x.median()) function is applied to each group to replacing missing values with the median of non-missing values within the group.

For MasVnrArea and GarageYrBlt, I filled the zero in the place of missing values. So, it can show No Garage or Masonry veneer area.

After filling and transforming the missing values I checked the missing values again. And this time no missing value was found. After doing all the preprocessing and handling missing values, now it’s time to Split the data and train it for the prediction.

Chapter 5 – Model Building and Visualization

Model evaluation and visualization are crucial steps in assessing the performance of machine learning models and gaining insights from their predictions. We split the data into training and testing sets, train the model, and evaluate its performance using metrics like accuracy and ROC curves. Visualization techniques such as confusion matrices, feature importance plots, and learning curves help us understand the model's behavior, identify patterns, and make informed decisions about model improvements. By combining evaluation metrics with visualizations, we gain a comprehensive understanding of the model's strengths and weaknesses, enabling us to refine the model and effectively communicate its results.

5.1 Data Splitting and Standardization

First, Let's do the data Split for training the model. I divided the dataset into two sets: Training set and Validation set and the ratio of 70 :30.

```
from sklearn.model_selection import train_test_split

# Split data into train and validation sets
X_train, X_test, y_train, y_test = \
    train_test_split(model_data.copy(), y, test_size=0.3, random_state=42)

# Print shapes of train and validation sets
print('Shapes')
print('X_train:', X_train.shape)
print('X_val:', X_test.shape)
print('y_train:', y_train.shape)
print('y_val:', y_test.shape)
```

```
Shapes
X_train: (1019, 263)
X_val: (437, 263)
y_train: (1019,)
y_val: (437,)
```

Fig. 22 – Data Split Code Snippet

As you can see, the **train_test_split** function from the **sklearn.model_selection** module is employed for this purpose. The dataset, comprising feature data stored in **model_data** and corresponding target labels denoted by **y**, is split into two sets: **X_train** and **y_train** for training, and **X_test** and **y_test** for validation. The parameter **test_size=0.3** specifies that 30% of the data should be allocated for validation, while **random_state=42** ensures reproducibility by fixing the random seed. This procedure enables the model to be trained on one portion of the data and evaluated on another, facilitating an objective assessment of its performance.

```
from sklearn.preprocessing import RobustScaler, StandardScaler

# Standardize numerical features using StandardScaler
stdsc = StandardScaler()
X_train.loc[:, num_features] = stdsc.fit_transform(X_train[num_features])
X_test.loc[:, num_features] = stdsc.transform(X_test[num_features])
```

Fig 23 – Standardize numerical features Code Snippet

You can see that, the training and testing datasets (X_train and X_test) are then standardized using the StandardScaler object. Standardization ensures that the features have a mean of zero and a standard deviation of one, which can improve the performance of certain machine learning algorithms.

```
from sklearn.model_selection import cross_val_score

def rmse(model, X, y):
    # Calculate root mean squared error using cross-validation
    cv_scores = -cross_val_score(model, X, y, scoring='neg_mean_squared_error', cv=10)
    return np.sqrt(cv_scores)
```

Fig. 24 – RSME Code Snippet

Then, the root mean squared error (RMSE) will be calculated for a given model using cross-validation. This function takes the model, feature matrix (X), and target variable (y) as inputs and computes the RMSE by performing cross-validation with 10 folds. Cross-validation is a technique used to assess the performance of a model on unseen data by splitting the dataset into multiple subsets, training the model on different subsets, and evaluating its performance on the remaining data. The RMSE is a common metric for regression tasks, measuring the average difference between the predicted and actual values, with lower values indicating better performance. Also, I used the R-squared score for more understanding of the model's accuracy. The higher R2 scores indicate superior model performance.

5.2 Model Evaluation and Visualization:

After the preprocessing, data splitting, and standardization, it’s time for Model training and Evaluation. Firstly, I tried to evaluate the performance of a Ridge regression model using GridSearchCV to find the optimal alpha value, which is a hyperparameter that controls the regularization strength and then refined the search around the optimal alpha values as shown in Fig. 24.

```
# Use GridSearchCV to find the optimal alpha value for Ridge regression
param_grid = {'alpha': [0.01, 0.1, 1., 5., 10., 25., 50., 100.]}
ridge = GridSearchCV(Ridge(), cv=5, param_grid=param_grid, scoring='neg_mean_squared_error')
ridge.fit(X_train, y_train)
alpha = ridge.best_params_['alpha']

# Refine the search around the optimal alpha value
param_grid = {'alpha': [x/100. * alpha for x in range(50, 150, 5)]}
ridge = GridSearchCV(Ridge(), cv=5, param_grid=param_grid, scoring='neg_mean_squared_error')
ridge.fit(X_train, y_train)
alpha = ridge.best_params_['alpha']
ridge = ridge.best_estimator_

# Calculate RMSE
train_rmse = rmse(ridge, X_train, y_train)
test_rmse = rmse(ridge, X_test, y_test)

# Calculate R-squared
train_r2_ridge = r2_score(y_train, ridge.predict(X_train))
test_r2_ridge = r2_score(y_test, ridge.predict(X_test))
```

Fig. 25 – Ridge Model Code Snippet

```
Ridge -> Train R-squared: 0.94207 | Test R-squared: 0.90885
Ridge -> Train RMSE: 0.09624 | Test RMSE: 0.11628 | alpha: 21.25000
```

Based on the output, the Root Mean Squared Error (RMSE) for the test dataset is approximately 0.11628, while for the training dataset, it's around 0.09624. These values suggest that the model's predictions are quite accurate, with minimal error between the predicted and actual values.

Additionally, the R-squared (R2) values for the test dataset and training dataset are approximately 0.90885 and 0.94, respectively. These high R2 values indicate that a significant proportion of the variance in the target variable is explained by the model, demonstrating its effectiveness Fig. 26.

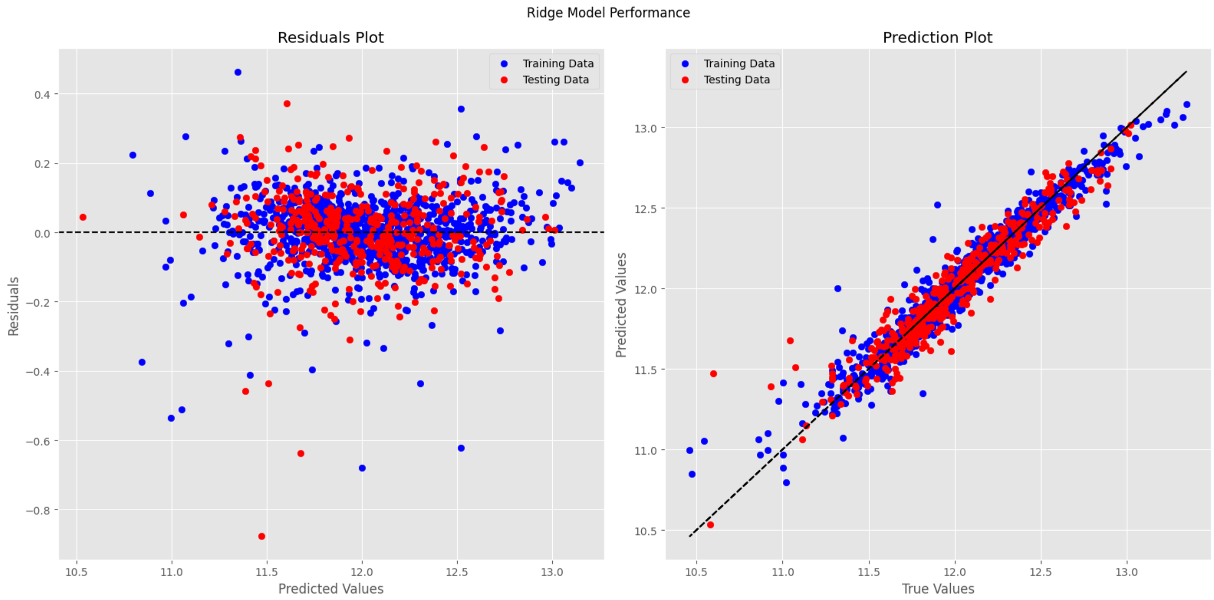


Fig. 26 - Ridge Model Performance

Also, I implemented a Lasso regression model using a similar approach. Subsequently, I refined the search around the optimal alpha value to further evaluate the model's performance. This iterative process allowed me to fine-tune the model and assess its efficiency.

The output I got from Lasso regression model is almost same as Ridge regression model.

```
Lasso -> Train RMSE: 0.09692 | Test RMSE: 0.11432 | alpha: 0.00055
Lasso -> Train R-squared: 0.94126 | Test R-squared: 0.91190
```

Based on the output, the Root Mean Squared Error (RMSE) of Lasso for the test dataset is approximately 0.11432, which is almost similar to the Ridge model 0.11628, while for the training dataset, it's around 0.9692 which is similar to the Ridge Model 0.09624. These values suggest that both models performance is almost similar to each other.

Additionally, the R-squared (R2) values are almost similar but the Lasso model is a bit better than the Ridge model. training dataset are approximately 0.91190 and 0.94126, respectively.

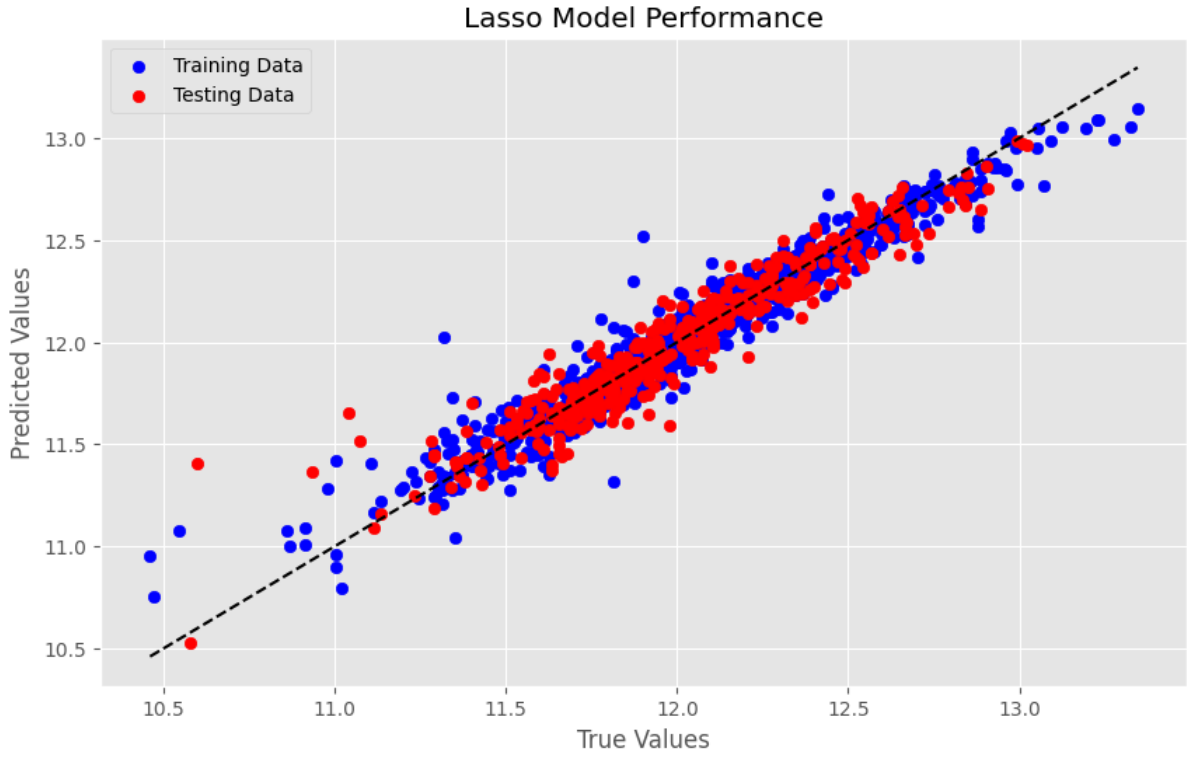


Fig. 27 – Lasso Model Performance

Talking about the remaining three models:

The Support Vector Regression model is initialized and fitted using the training data. Then, predictions are made for both the training and testing datasets. The R-squared (R2) values are predicted:

```
SVR -> Train R-squared: 0.96447 | Test R-squared: 0.87208
```

And the model calculates the Root Mean Squared Error (RMSE) predictions for test and train data are:

```
SVR -> Train RMSE: 0.07537 | Test RMSE: 0.13776
```

We can see that the performance of this model is not better if we compare it to the above two models, where R-squared test values are above 0.90 as we compare with the SVR model which is 0.87208, which Is not good enough.

Also, the RMSE of the previous two models is much better than this model. Where Test RMSE is almost 0.11 in both models, now it 0.13776.

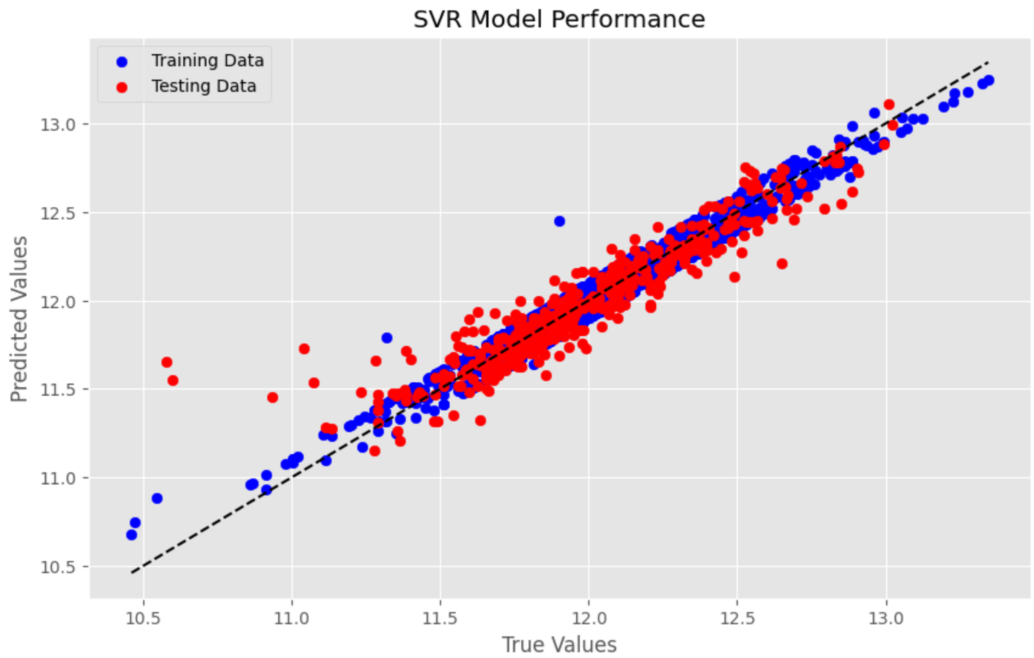


Fig. 28 – SVR model Performance

Also, For Random Forest, I got almost similar outputs compared to the SVR model.

```
Random Forest -> Train R-squared: 0.98359 | Test R-squared: 0.87314
Random Forest -> Train RMSE: 0.05123 | Test RMSE: 0.13719
```

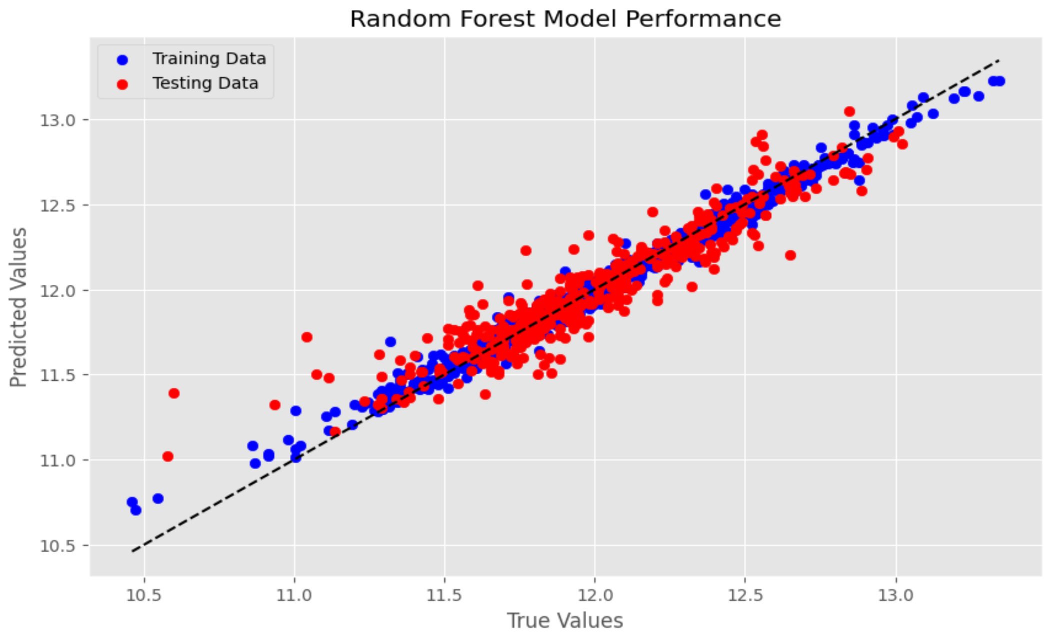


Fig. 29 – Random Forest Model Performance

5.3 Comparison of the Outputs:

Let’s compare all the model's output results with the help of Table 1.

Method	RMSE(Test)	R2(Test)
Lasso Regression	0.11432	0.91190
Ridge Regression	0.11628	0.90885
SVR	0.13776	0.87208
Random Forest	0.13719	0.87314

Table 1. Output Results Comparison

We can observe that both the Lasso and Ridge models performed similarly, yielding RMSE test data results of 0.11432 and 0.11628, respectively, and R2 test data results of 0.91190 and 0.90885. This suggests that both models are equally effective in predicting house prices based on the given features.

On the other hand, when comparing the SVR and Random Forest models, we see that they have slightly higher RMSE test data results of 0.13776 and 0.13719, respectively, and slightly lower R2 test data results of 0.87208 and 0.87314. These results indicate that the SVR and Random Forest models may not perform as well as the Lasso and Ridge models in terms of predictive accuracy.

One possible reason for the relatively poorer performance of the SVR and Random Forest models could be attributed to the lack of fine-tuning using techniques like GridSearchCV. Unlike the Lasso and Ridge models, which underwent a refinement process to optimize hyperparameters, the SVR and Random Forest models may not have been subjected to such rigorous optimization, leading to suboptimal performance as you can see in Fig. 30 and Fig.31.

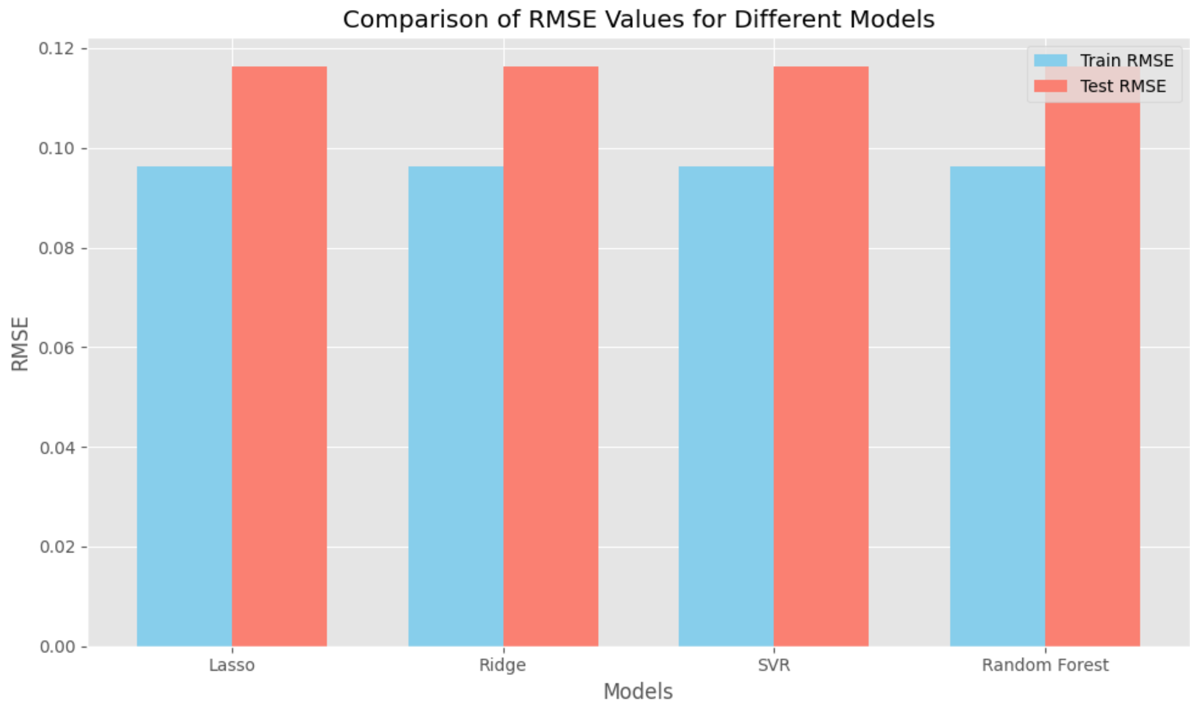


Fig 29 – RMSE Comparsion

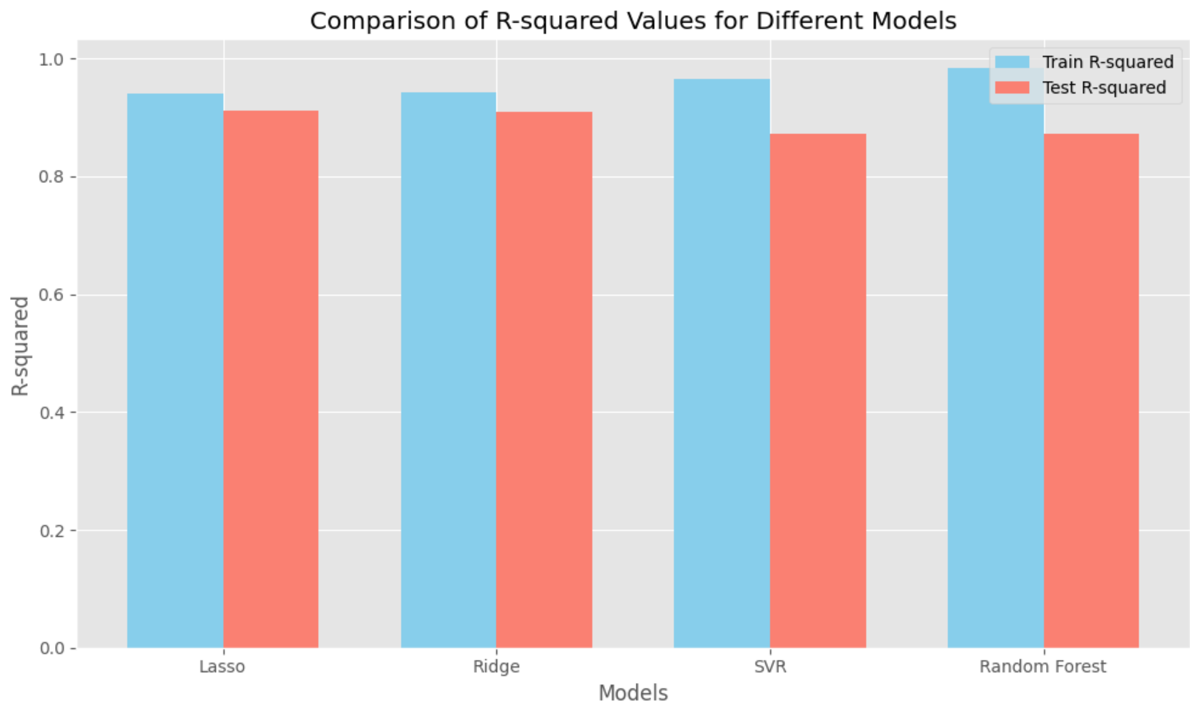


Fig. 31 – R-Squared Comparison

Furthermore, it's essential to consider the inherent differences in these models' algorithms and how they handle the data. For instance, SVR relies on support vectors to identify patterns in the data, while Random Forest uses an ensemble of decision trees. These differences may also contribute to variations in performance.

Chapter 6 – Conclusion

6.1 Conclusion:

In conclusion, this analysis provides valuable insights into the factors influencing residential property sale prices and the performance of various regression models in predicting these prices. Through exploratory data analysis, we gained a comprehensive understanding of the dataset's characteristics, revealing trends in house prices, construction years, sales patterns, and house sizes. Data preprocessing techniques were then applied to clean and transform the data, preparing it for modeling.

Subsequently, multiple regression models including Ridge, Lasso, SVR, and Random Forest were trained and evaluated for their predictive accuracy. The results indicate that both Ridge and Lasso regression models performed similarly well, with high R-squared values and relatively low RMSE values, demonstrating their effectiveness in predicting house prices. However, the SVR and Random Forest models exhibited slightly lower performance, possibly due to the lack of fine-tuning and optimization.

6.2 Future Work:

Moving forward, several avenues for future work present themselves. Firstly, further optimization and fine-tuning of models such as SVR and Random Forest could potentially improve their predictive accuracy. Techniques like hyperparameter tuning and ensemble methods could be explored to enhance model performance.

Additionally, incorporating external data sources such as economic indicators, neighborhood demographics, and property market trends could enrich the analysis and provide additional insights into house price dynamics. Feature selection techniques could also be employed to identify the most influential predictors of house prices, thereby refining the models and improving interpretability.

Furthermore, exploring advanced machine learning techniques such as gradient boosting and neural networks could offer new perspectives and potentially higher predictive accuracy. Finally, conducting longitudinal studies to track changes in house prices over time and assessing the robustness of predictive models in different market conditions could provide valuable insights for stakeholders in the real estate industry.

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